

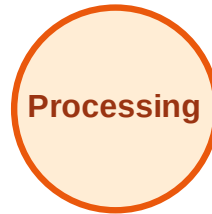
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

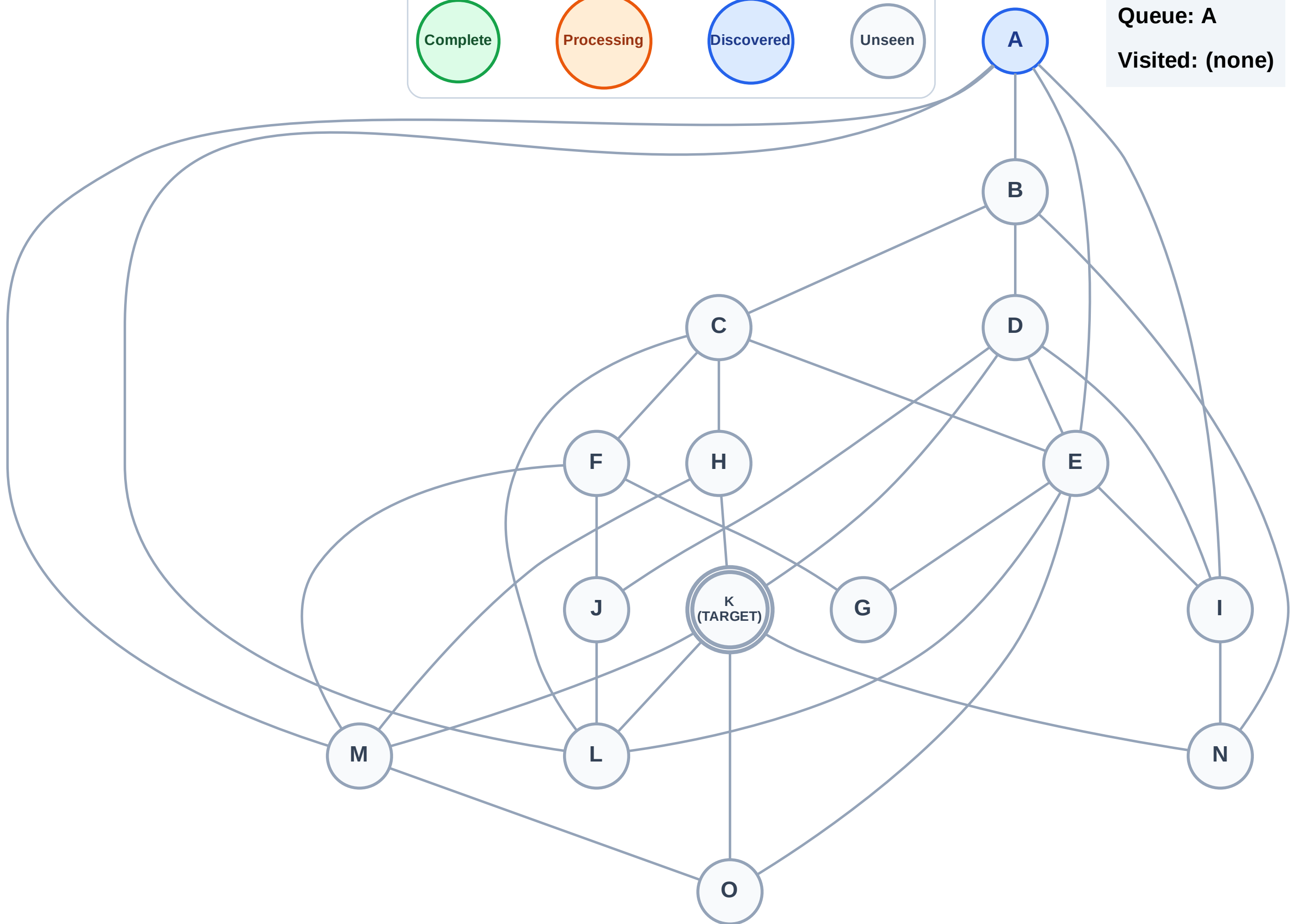


Discovered



Unseen

Queue: A
Visited: (none)



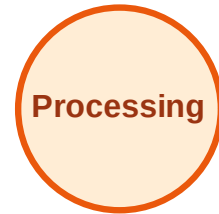
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



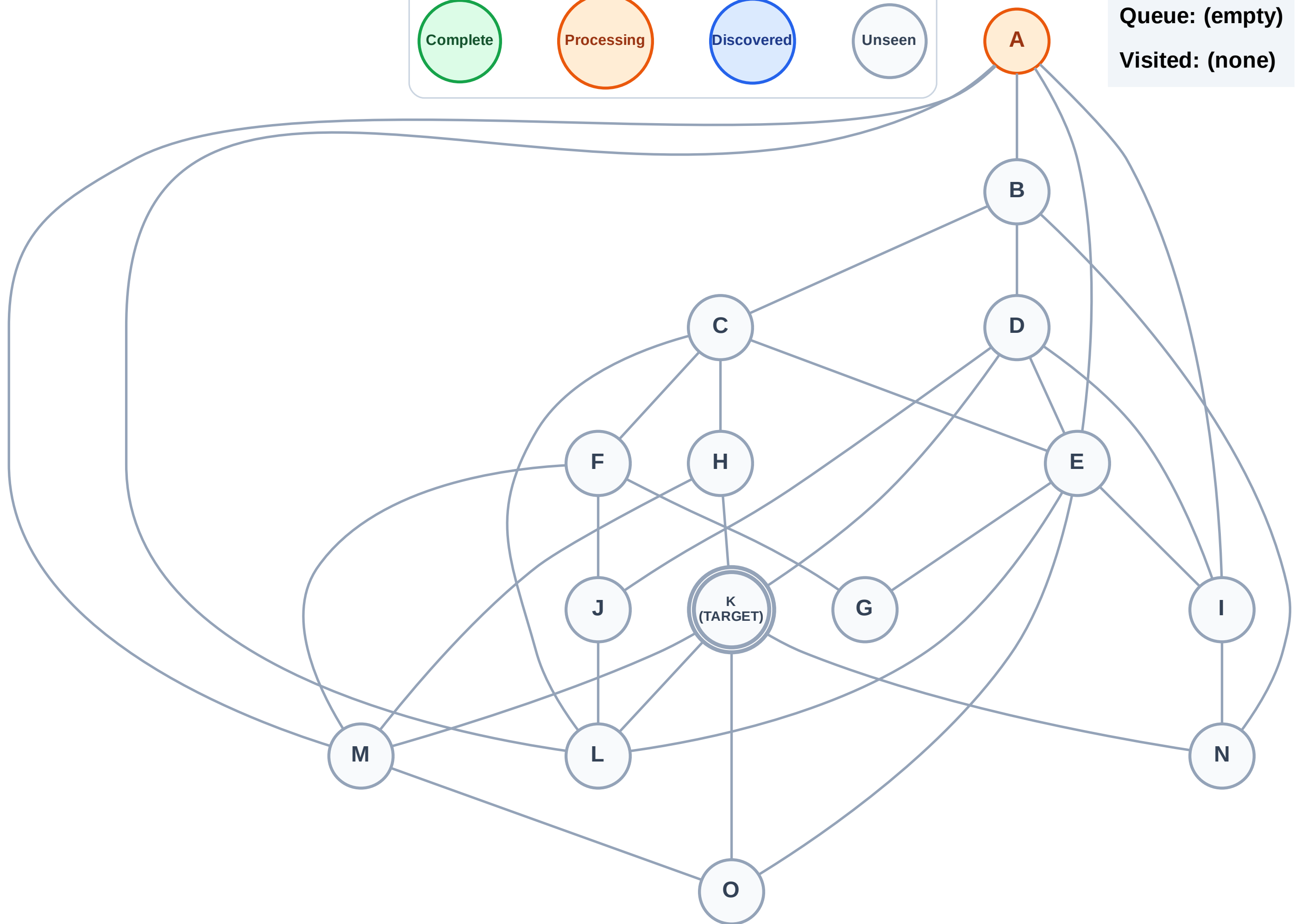
Discovered



Unseen

Queue: (empty)

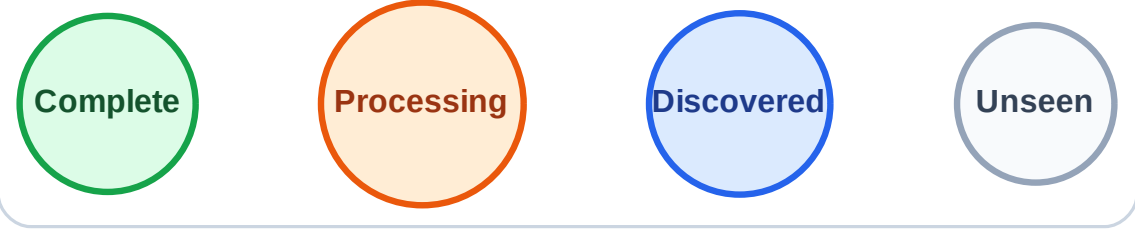
Visited: (none)



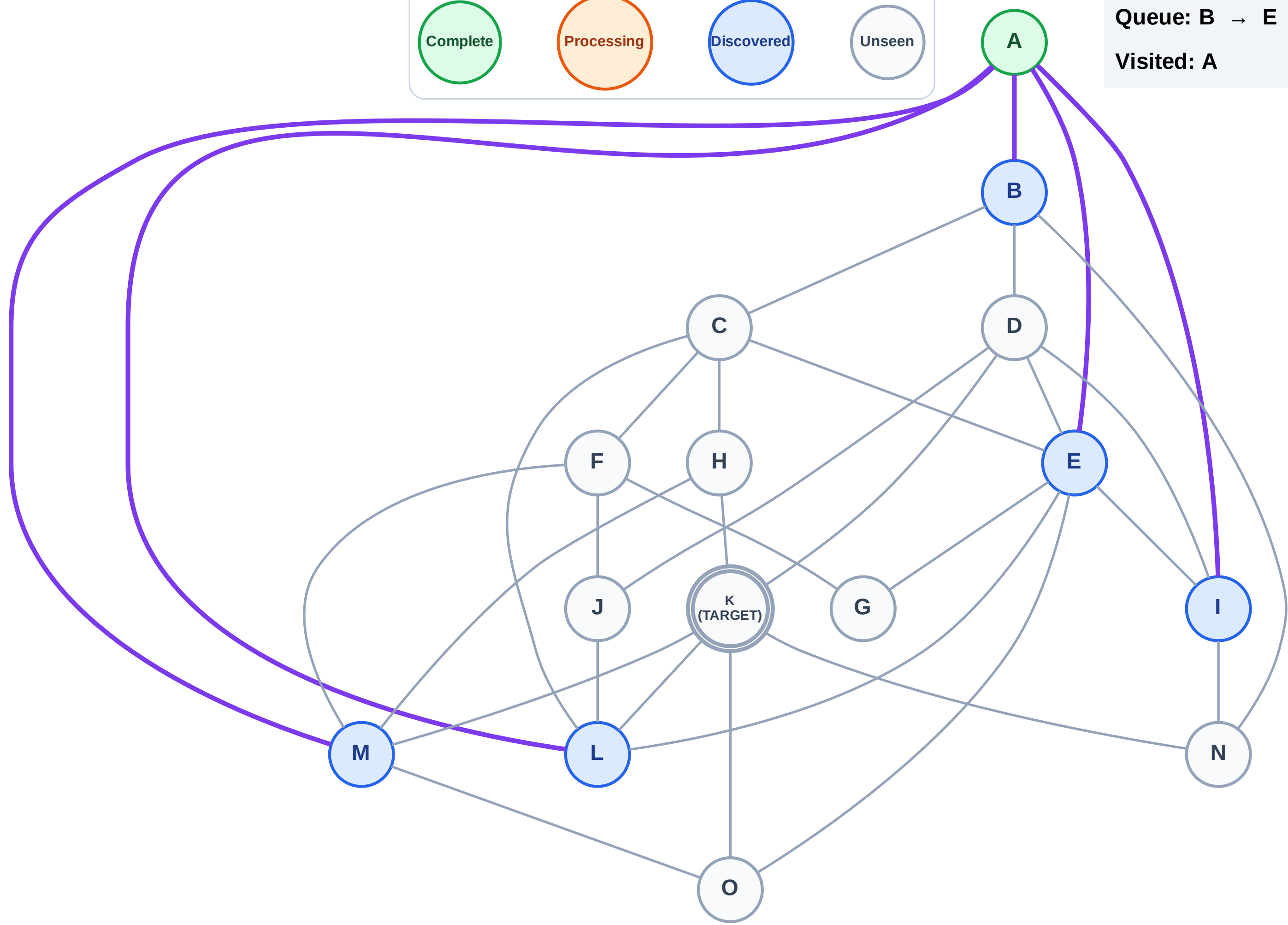
BFS Traversal

Step 3: Discover B, E, I, L, M from A

Legend



Queue: B → E → I → L → M
Visited: A



BFS Traversal

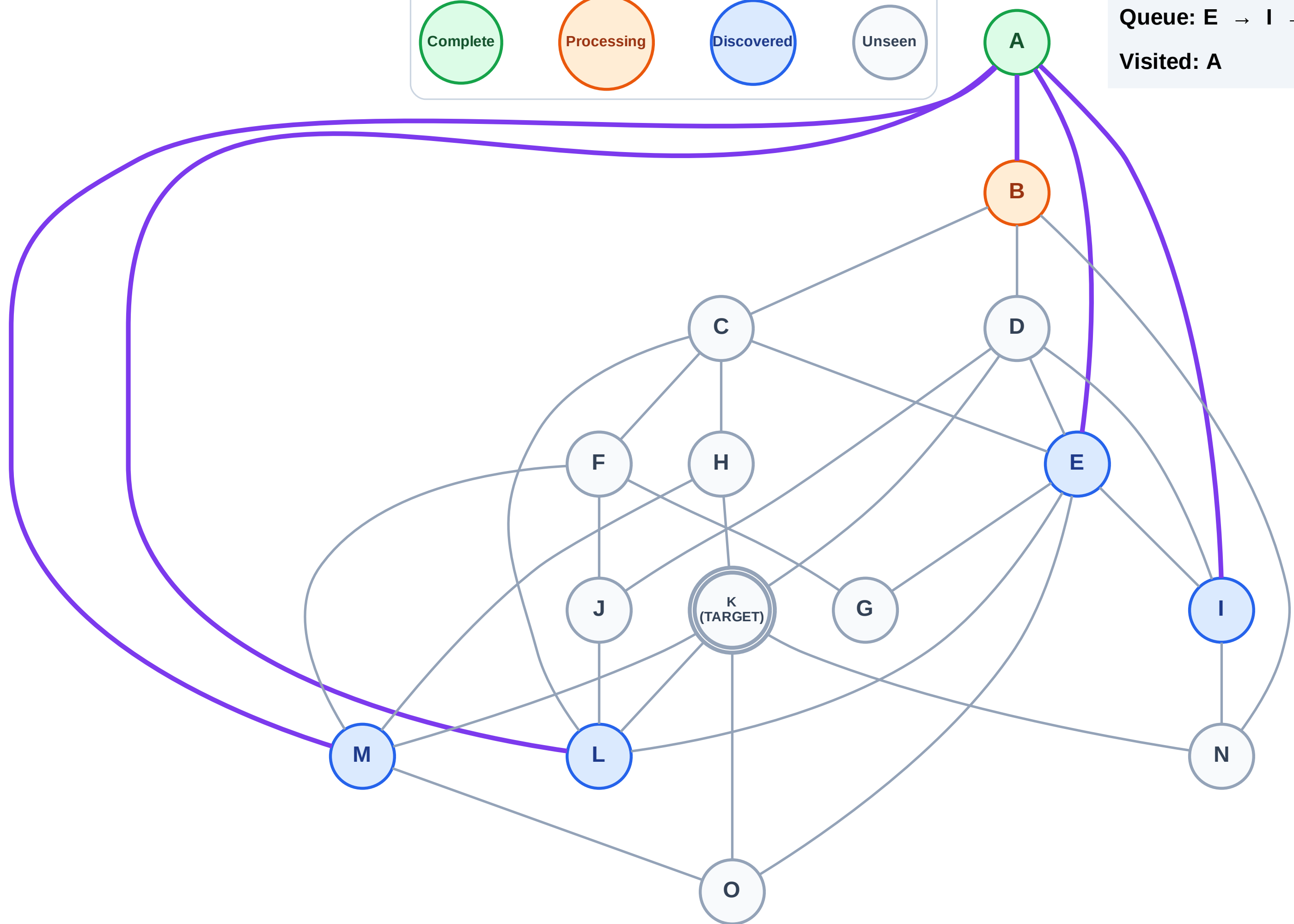
Step 4: Process node B

Legend



Queue: E → I → L → M

Visited: A



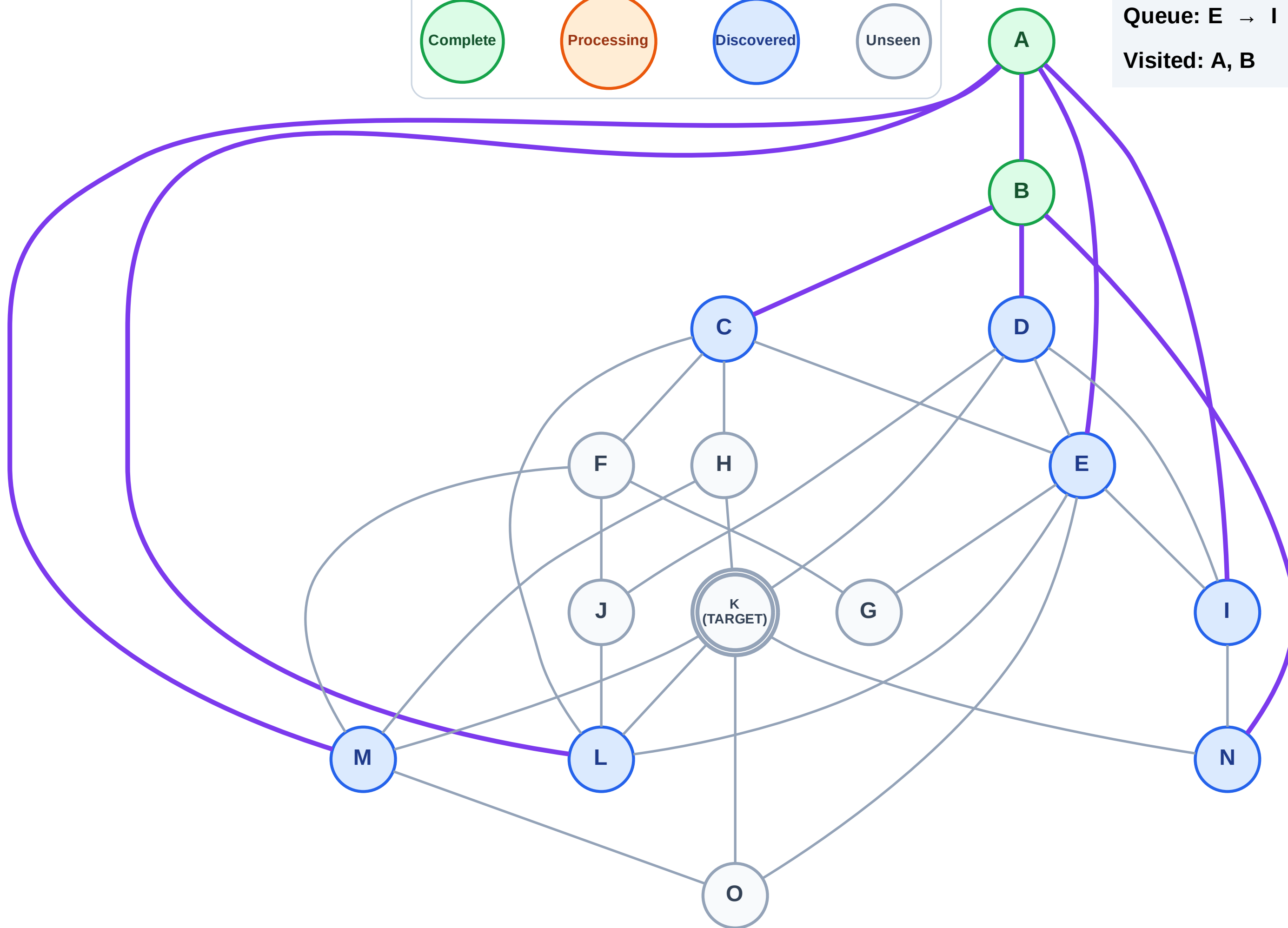
BFS Traversal

Step 5: Discover C, D, N from B

Legend



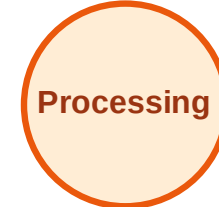
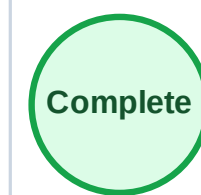
Queue: E → I → L → M → C → D → N
Visited: A, B



BFS Traversal

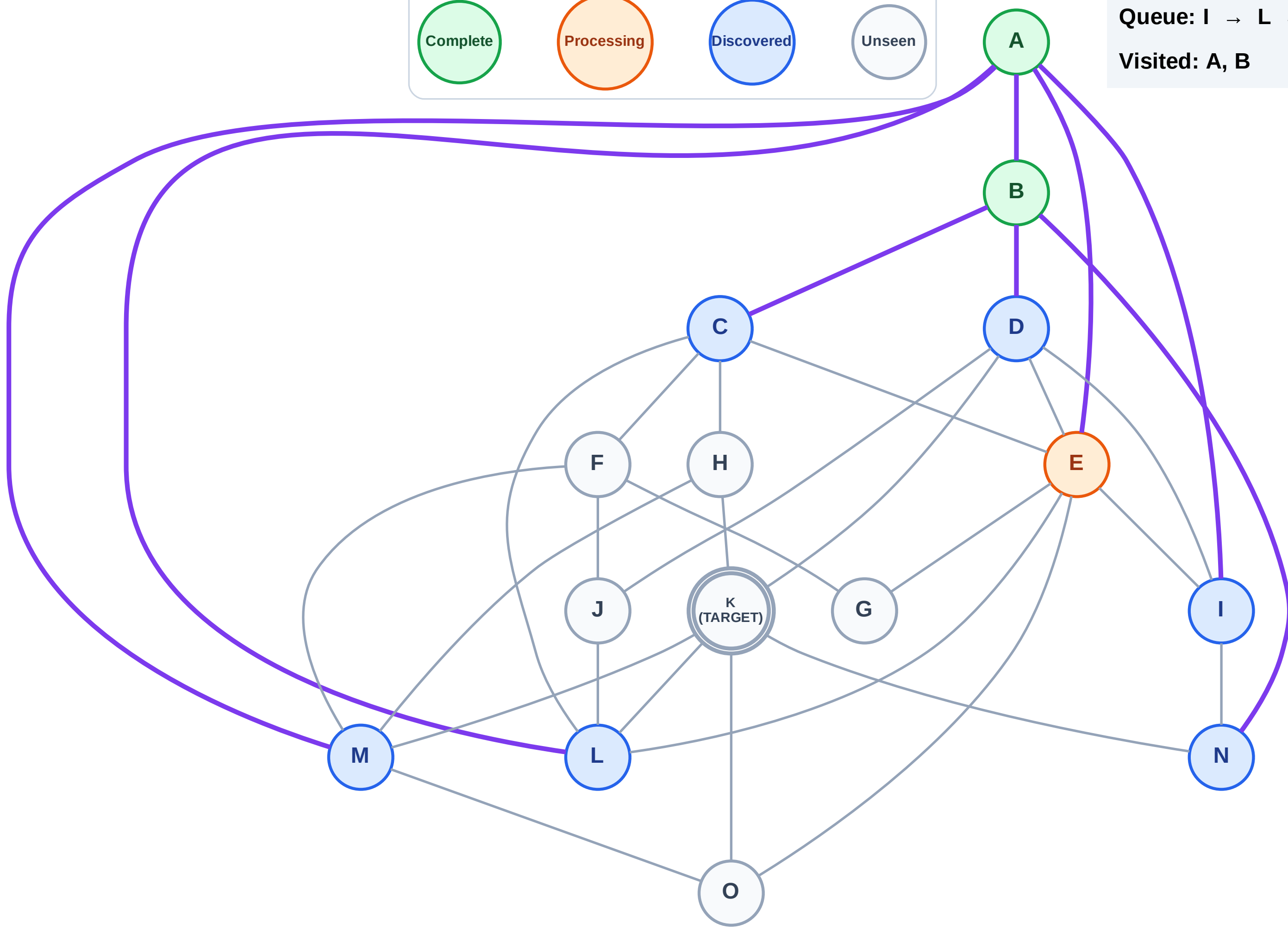
Step 6: Process node E

Legend



Queue: I → L → M → C → D → N

Visited: A, B



BFS Traversal

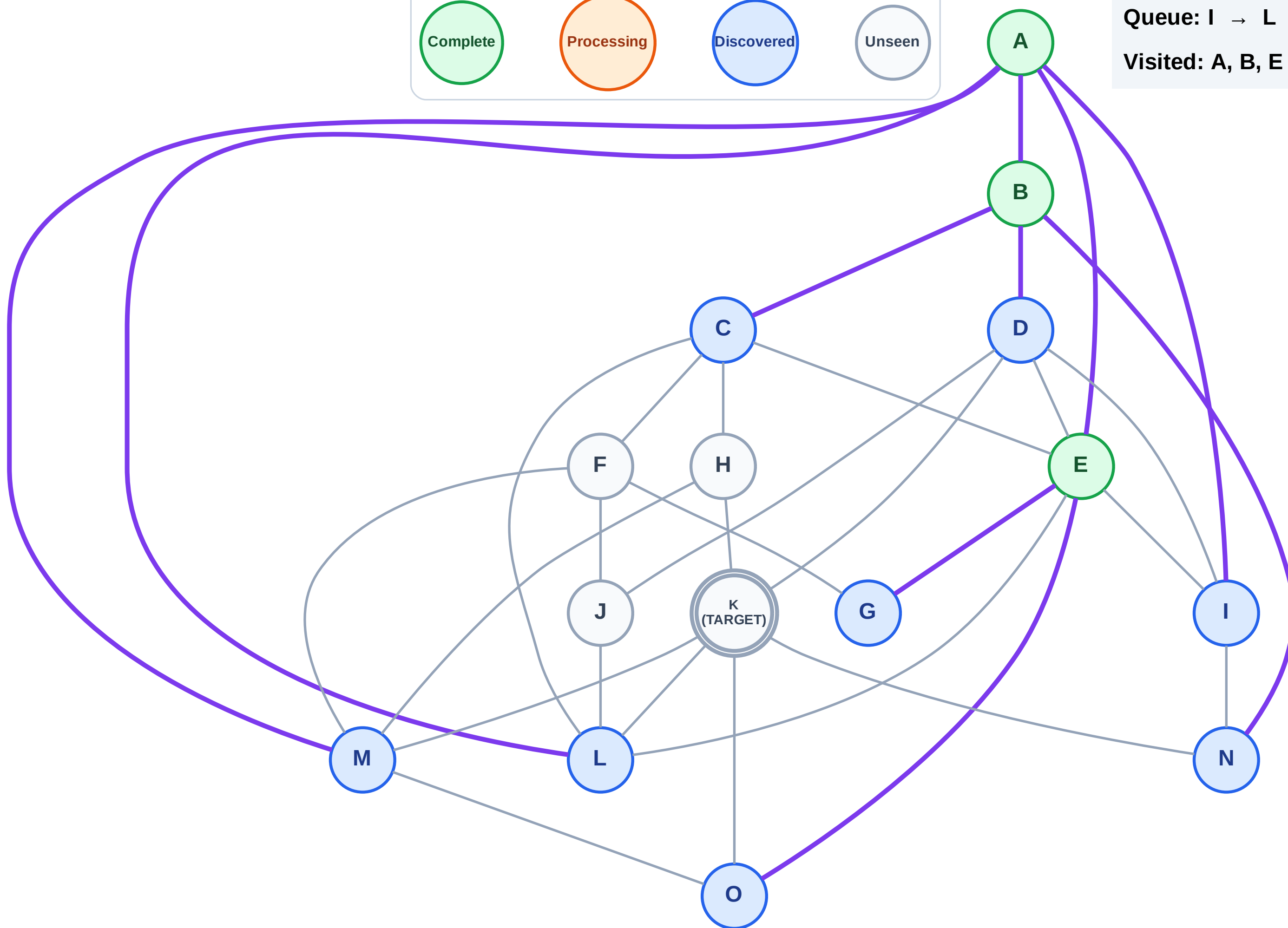
Step 7: Discover G, O from E

Legend



Queue: I → L → M → C → D → N → G → O

Visited: A, B, E



BFS Traversal

Step 8: Process node I

Legend

Complete

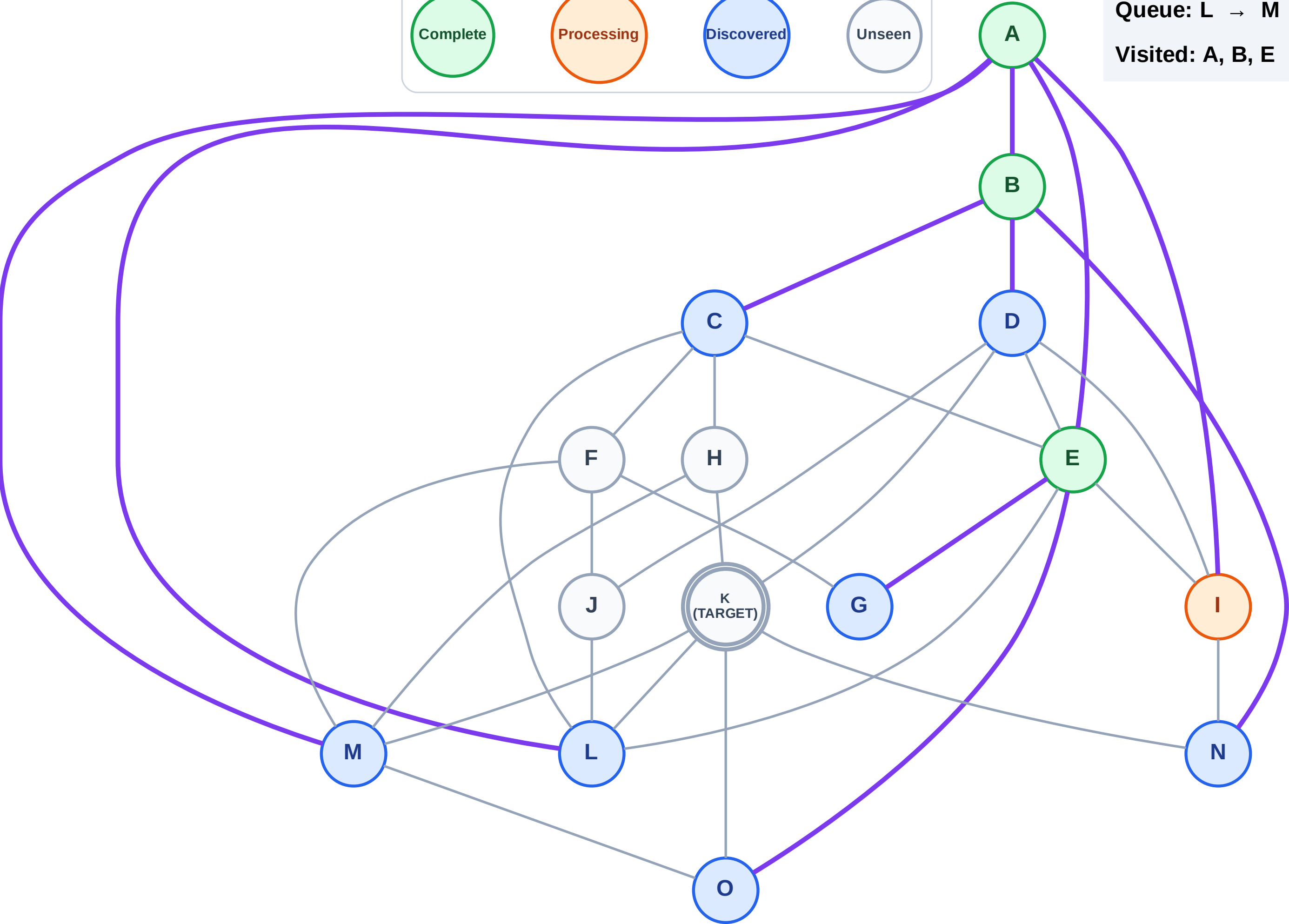
Processing

Discovered

Unseen

Queue: L → M → C → D → N → G → O

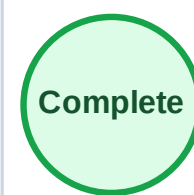
Visited: A, B, E



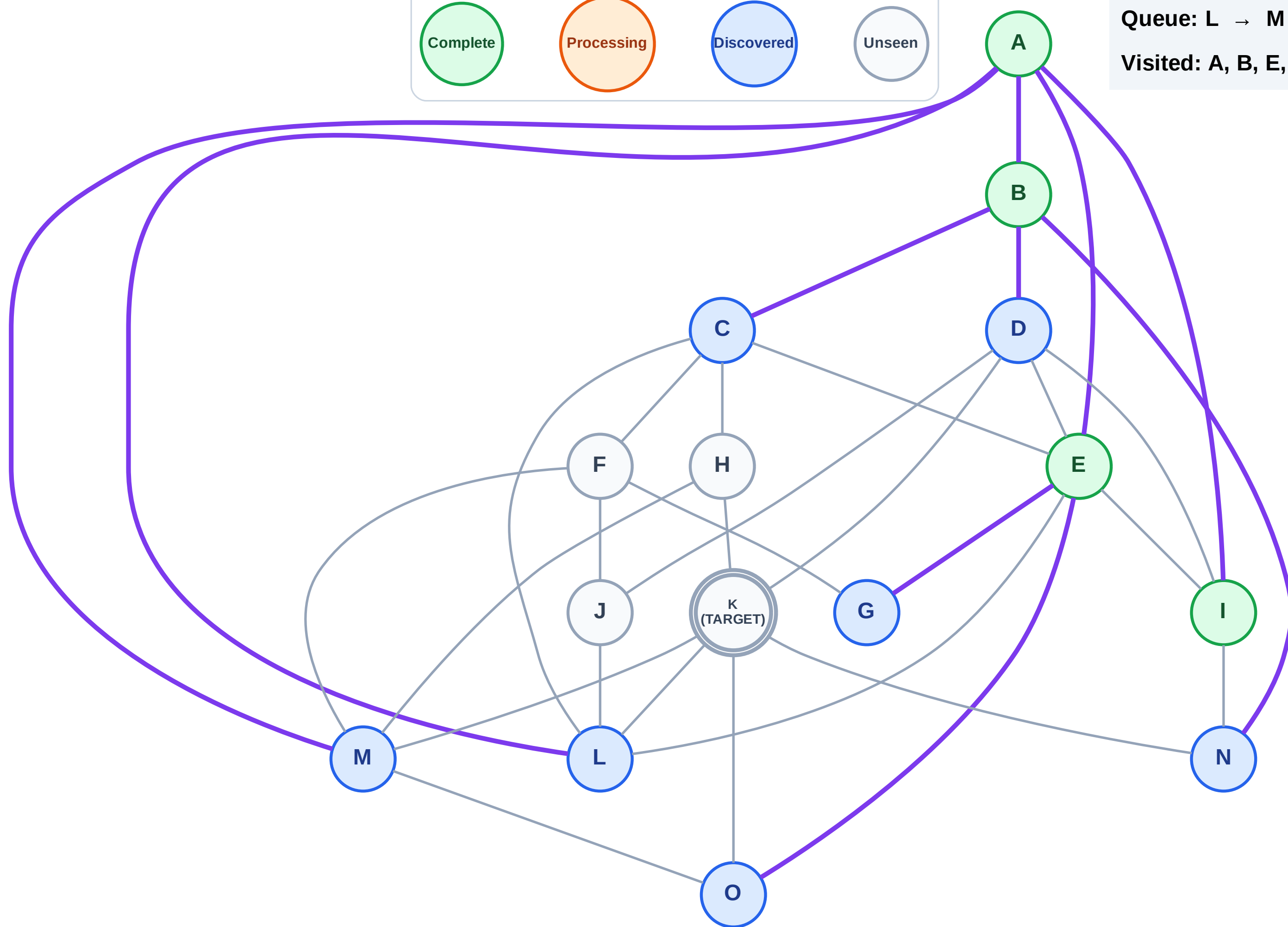
BFS Traversal

Step 9: No new neighbors from I

Legend



Queue: L → M → C → D → N → G → O
Visited: A, B, E, I



BFS Traversal

Step 10: Process node L

Legend



Complete



Processing



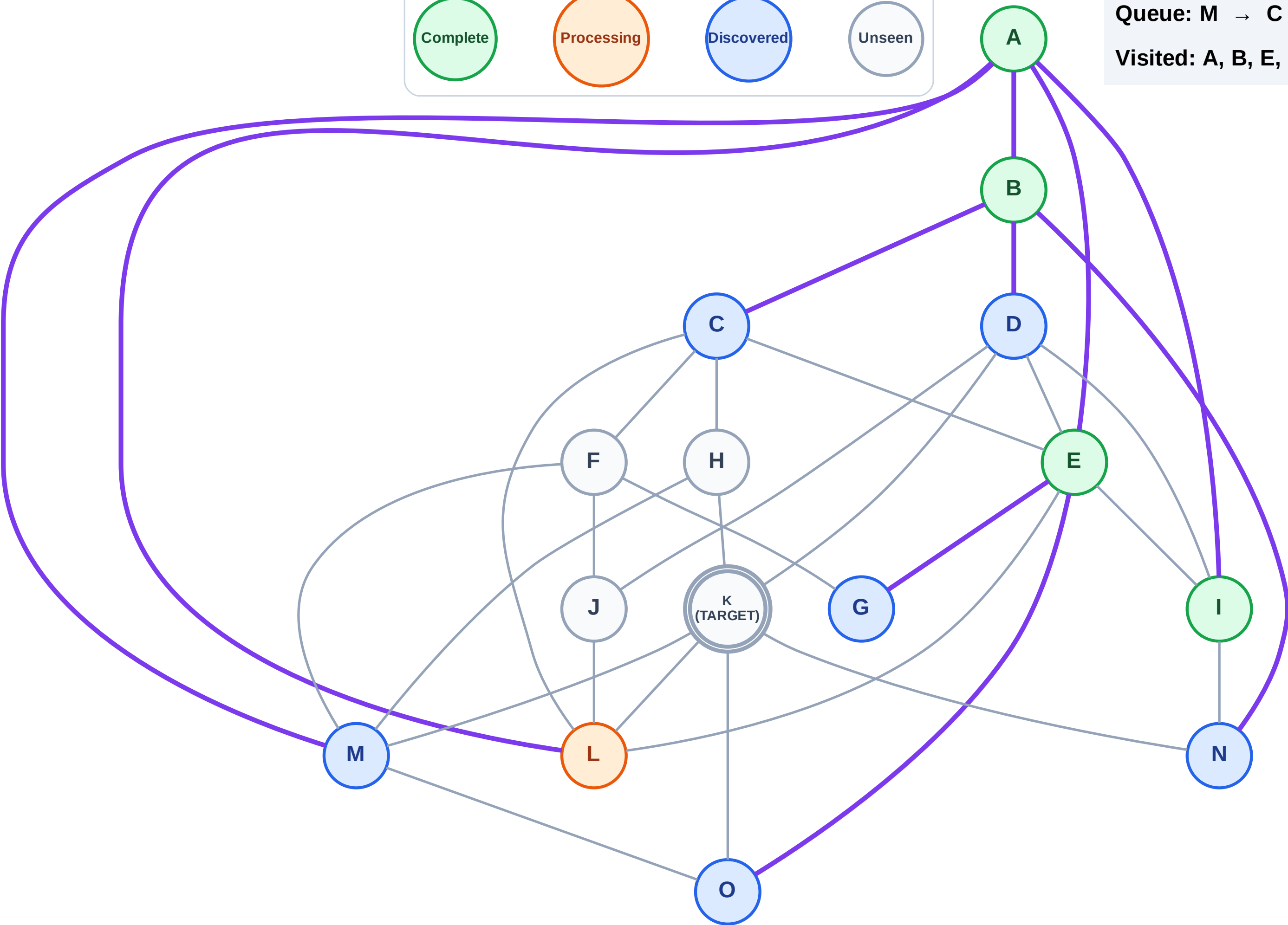
Discovered



Unseen

Queue: M → C → D → N → G → O

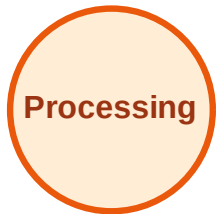
Visited: A, B, E, I



BFS Traversal

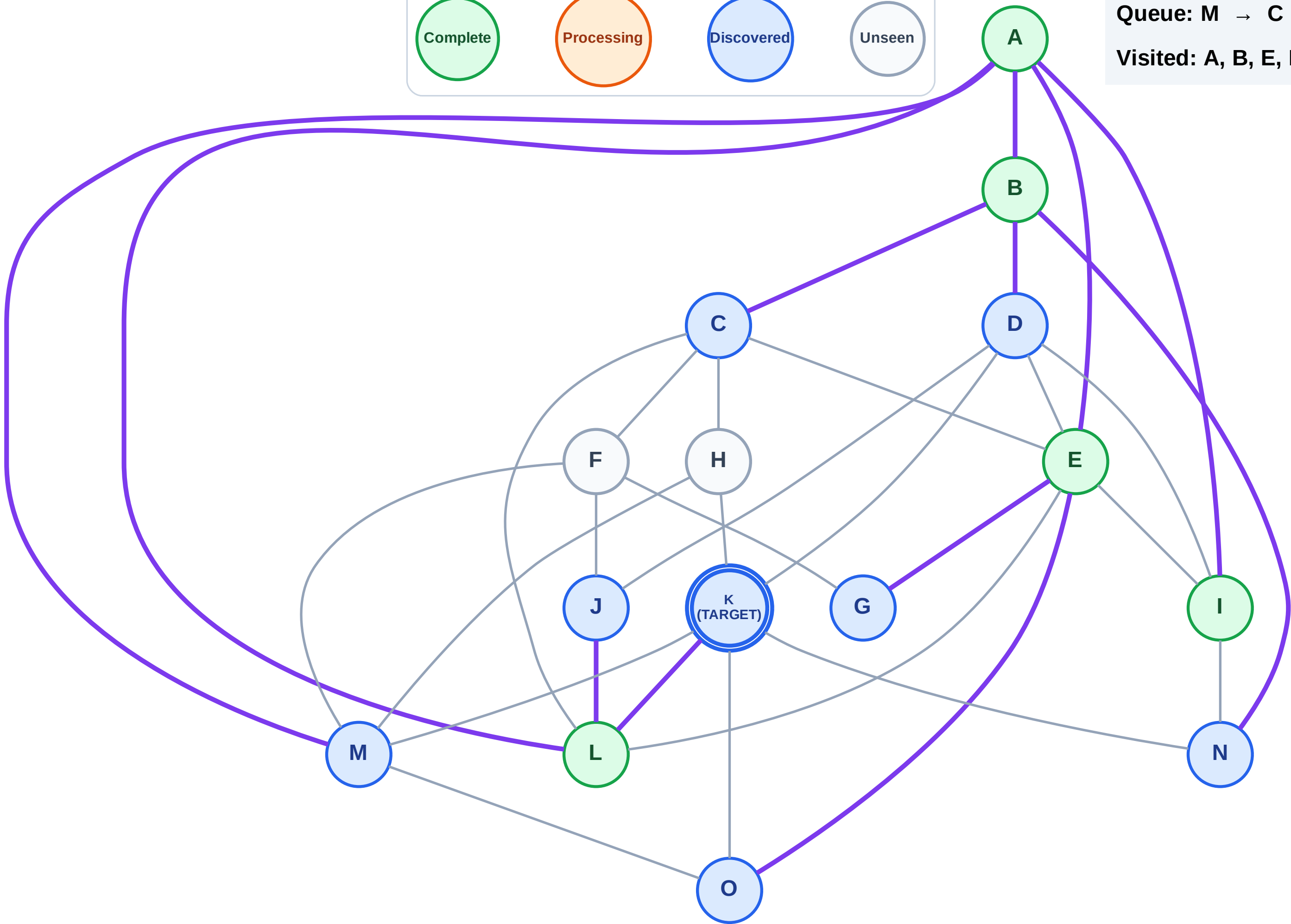
Step 11: Discover J, K from L

Legend



Queue: M → C → D → N → G → O → J → K

Visited: A, B, E, I, L



BFS Traversal

Step 12: Process node M

Legend

Complete

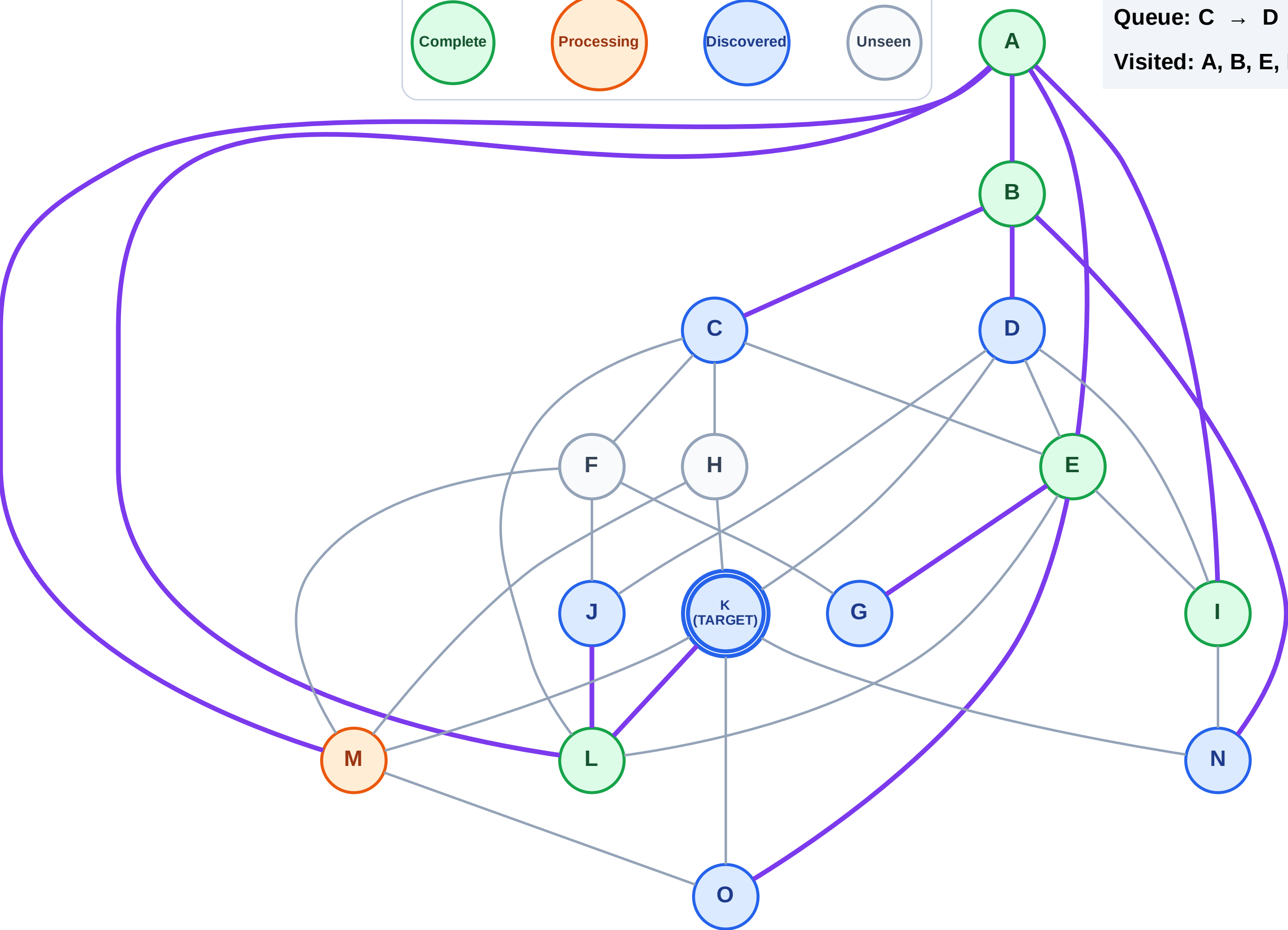
Processing

Discovered

Unseen

Queue: C → D → N → G → O → J → K

Visited: A, B, E, I, L



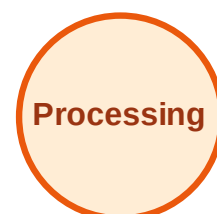
BFS Traversal

Step 13: Discover F, H from M

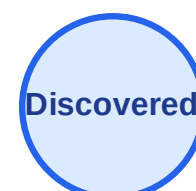
Legend



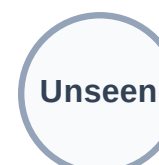
Complete



Processing



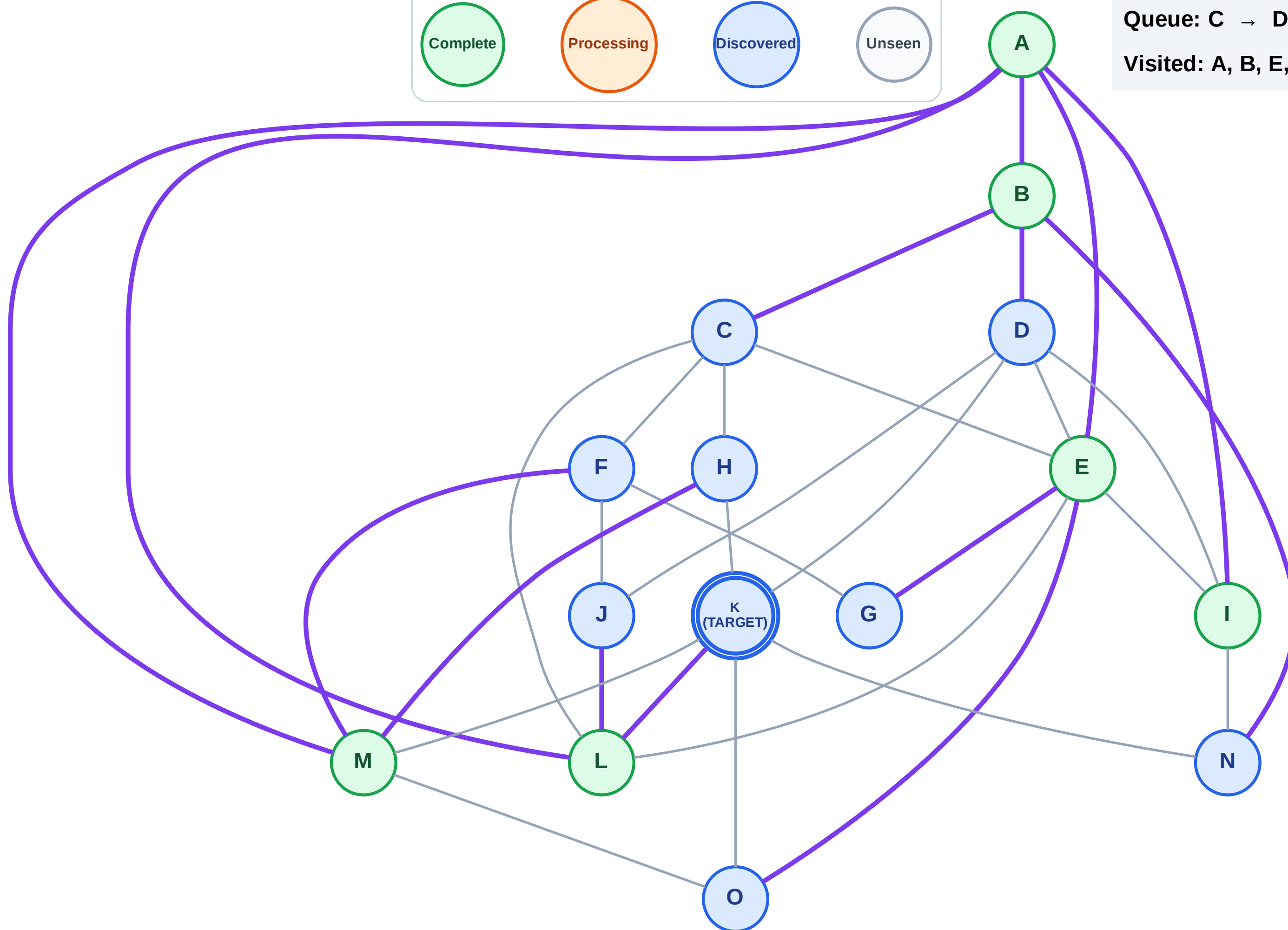
Discovered



Unseen

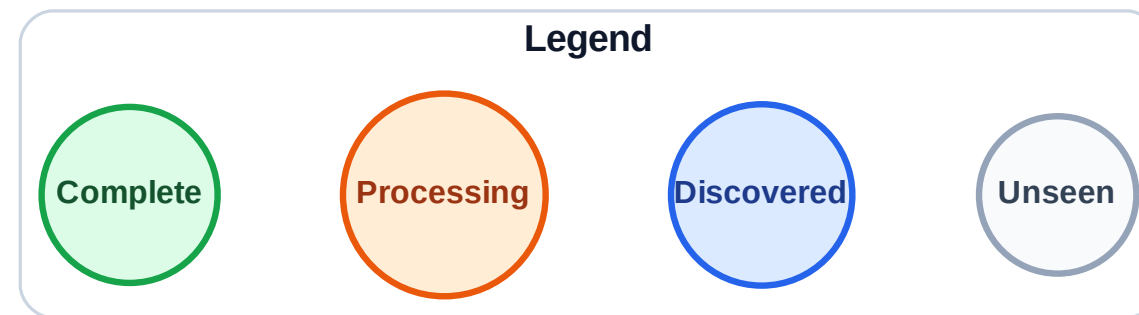
Queue: C → D → N → G → O → J → K → F → H

Visited: A, B, E, I, L, M

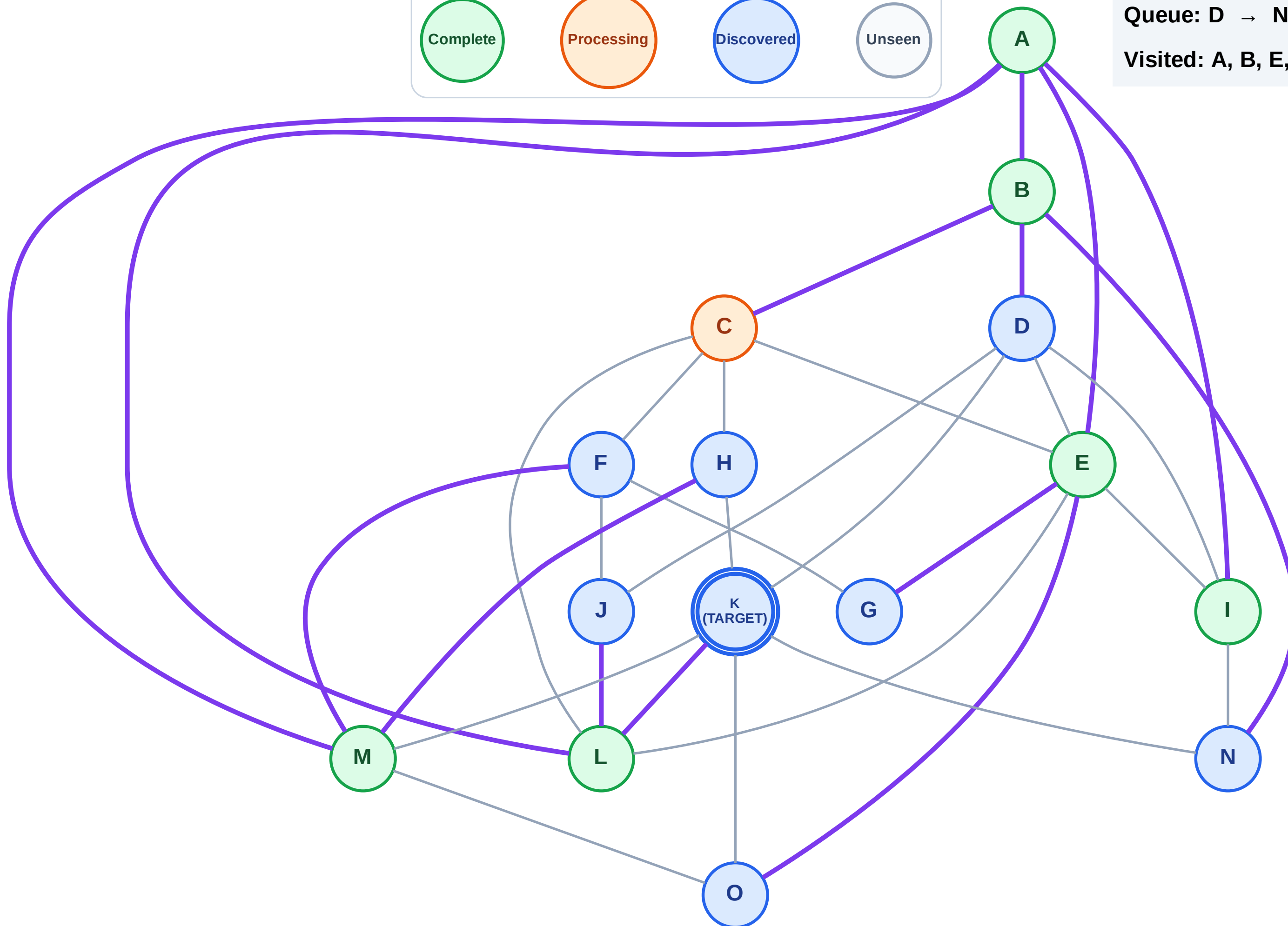


BFS Traversal

Step 14: Process node C



Queue: D → N → G → O → J → K → F → H
Visited: A, B, E, I, L, M



BFS Traversal

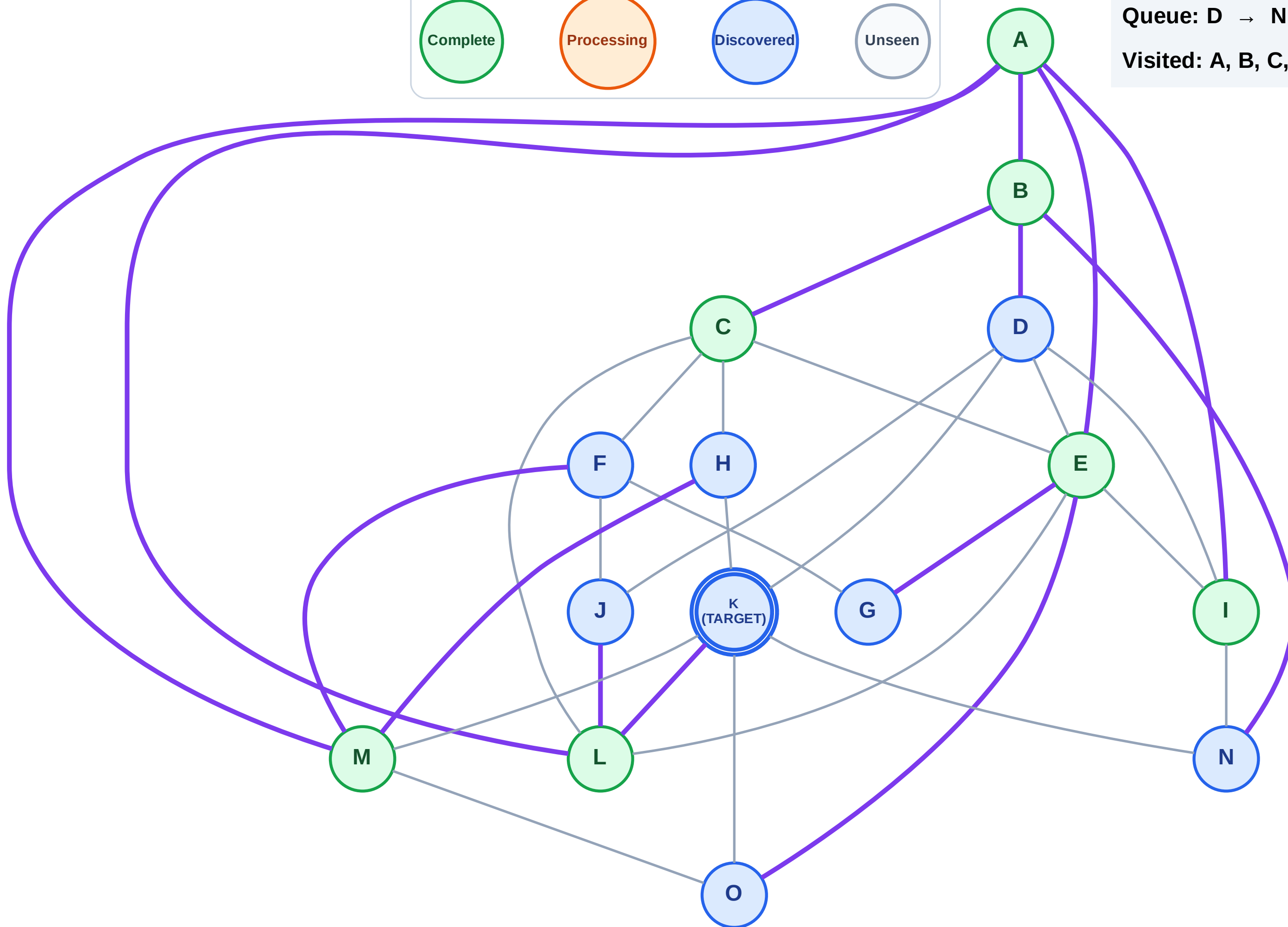
Step 15: No new neighbors from C

Legend



Queue: D → N → G → O → J → K → F → H

Visited: A, B, C, E, I, L, M

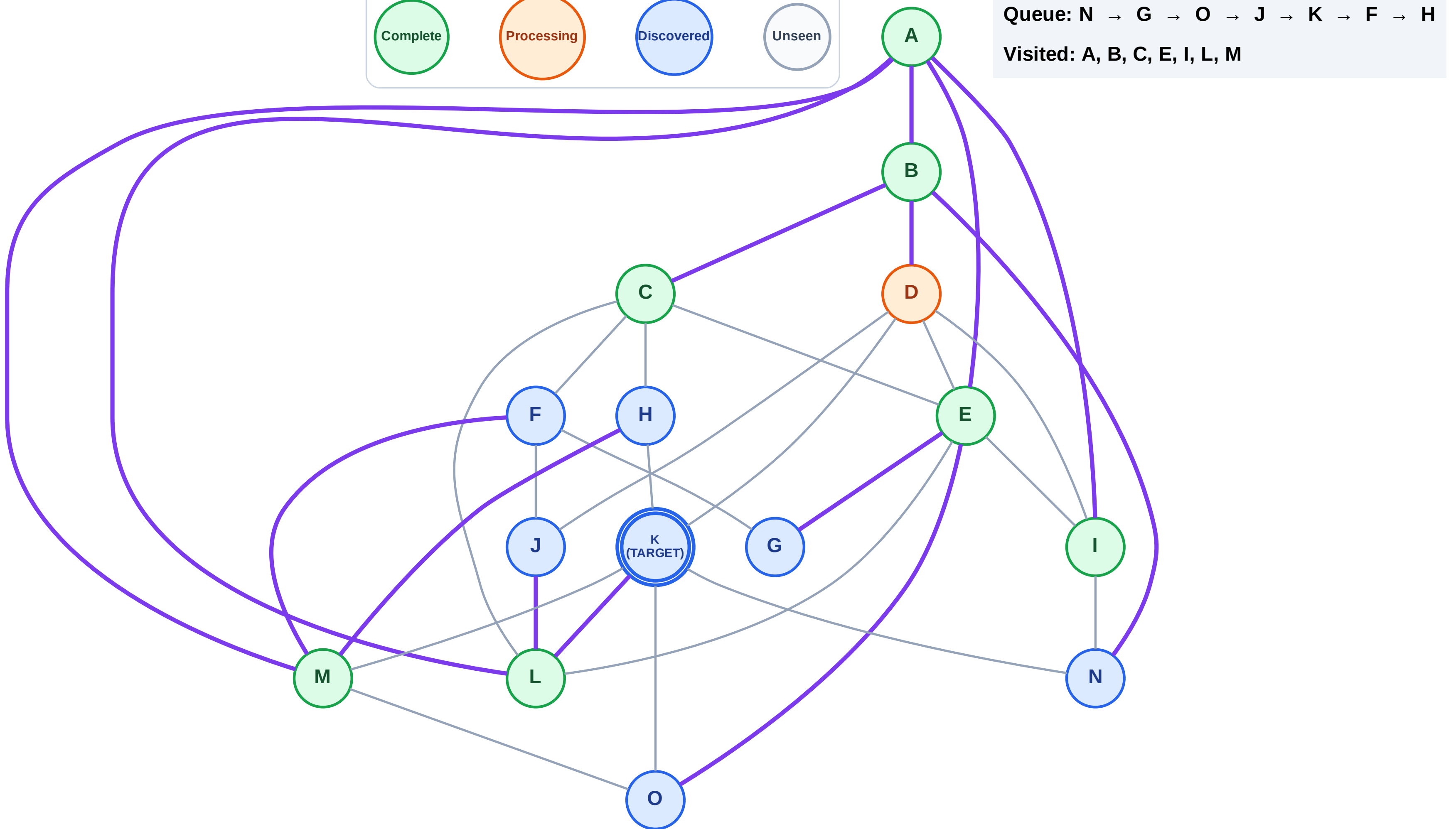


Step 16: Process node D

Complete Processing Discovered Unseen

Unseen

Visited: A, B, C, E, I, L, M



BFS Traversal

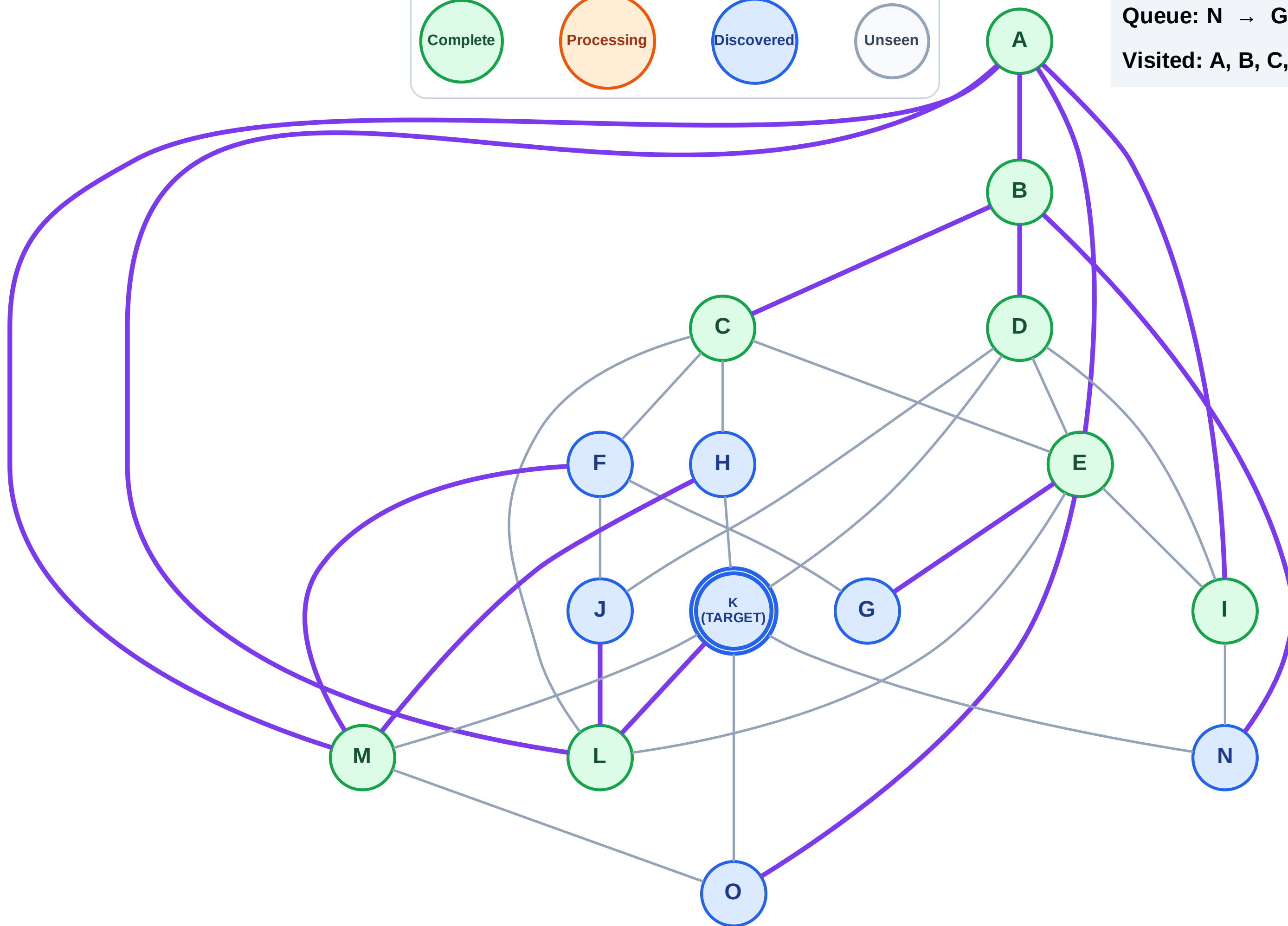
Step 17: No new neighbors from D

Legend



Queue: N → G → O → J → K → F → H

Visited: A, B, C, D, E, I, L, M



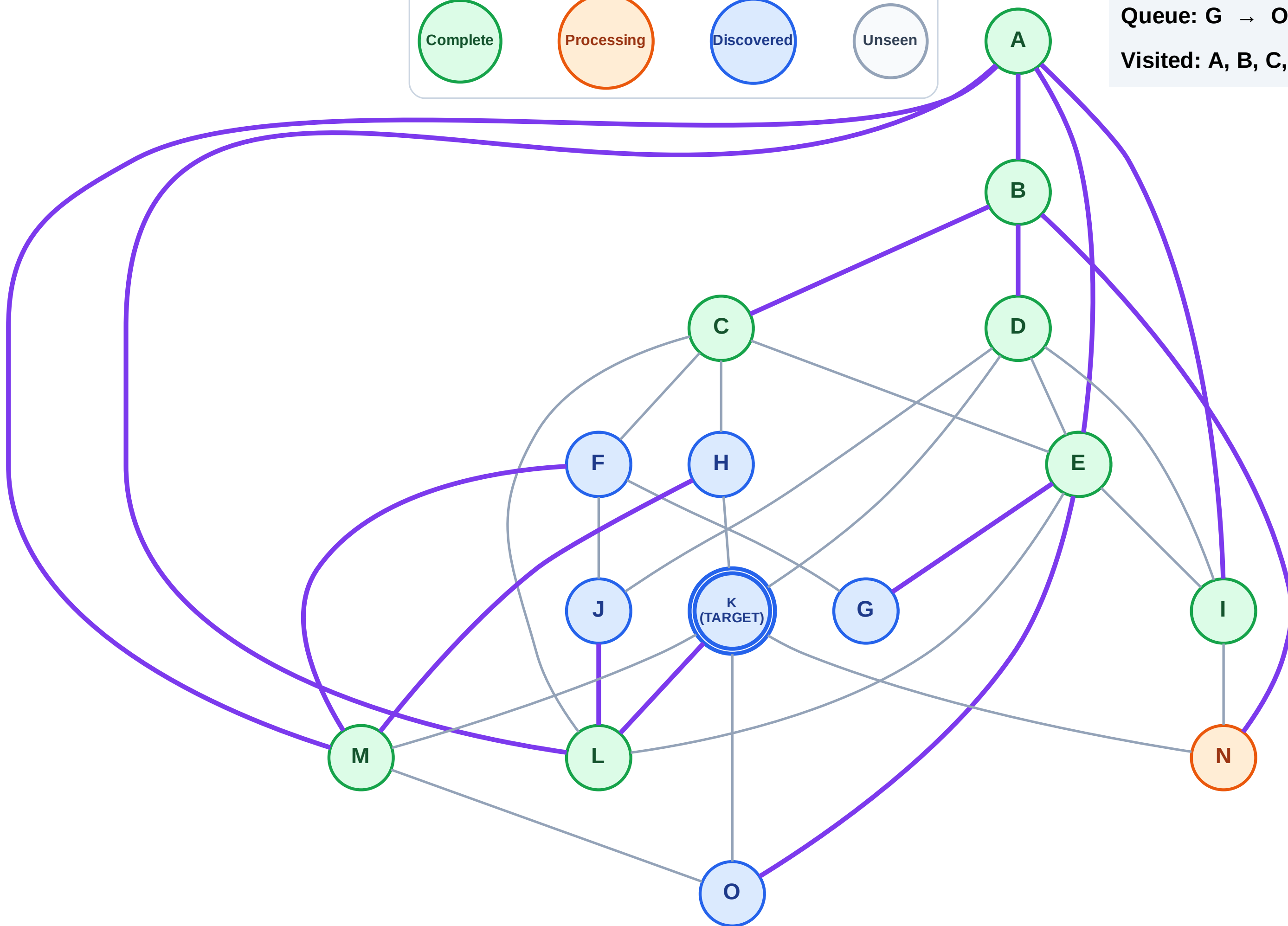
BFS Traversal

Step 18: Process node N

Legend



Queue: G → O → J → K → F → H
Visited: A, B, C, D, E, I, L, M



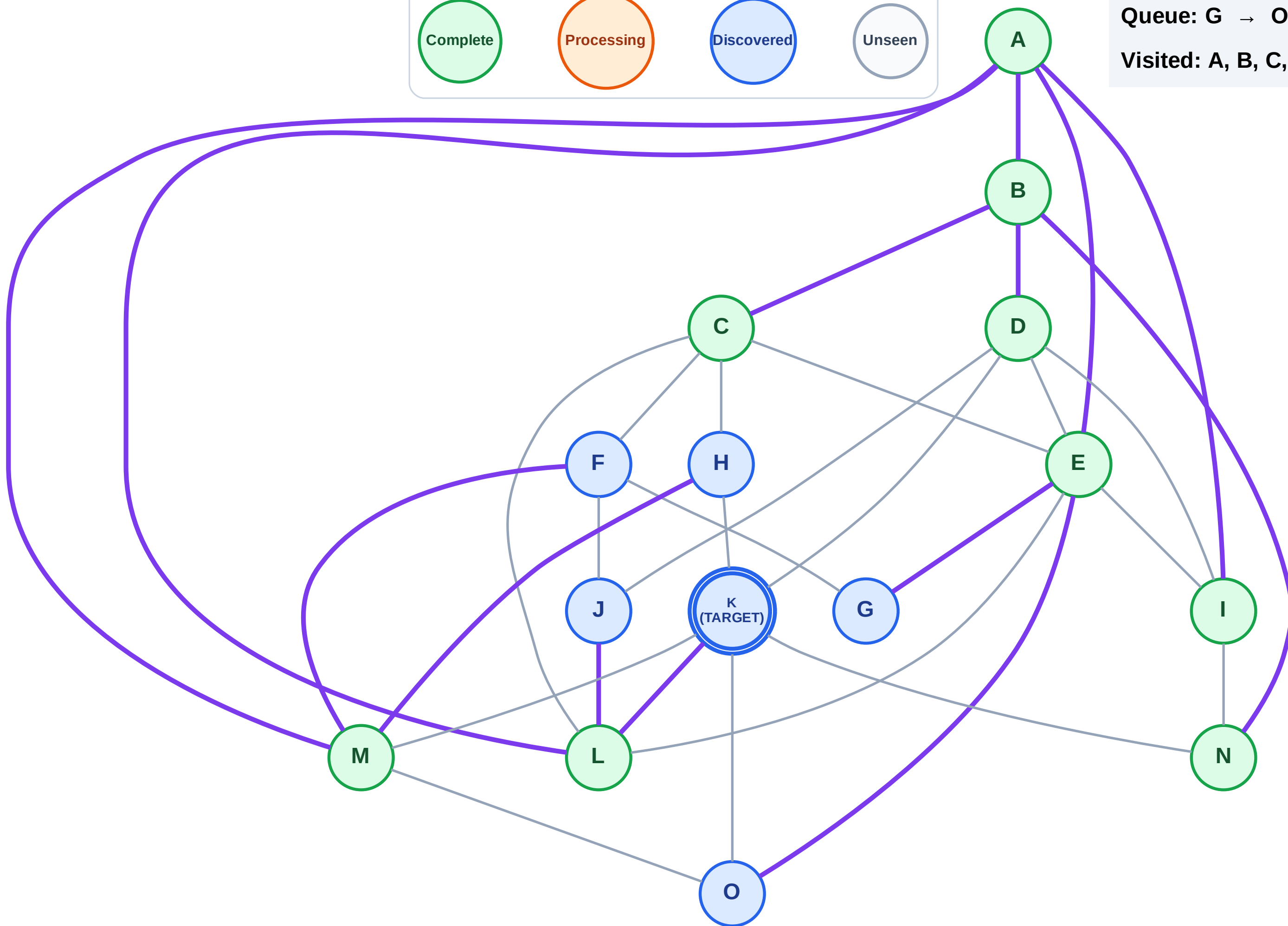
BFS Traversal

Step 19: No new neighbors from N

Legend



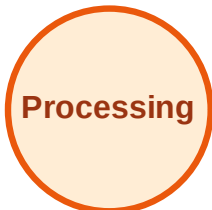
Queue: G → O → J → K → F → H
Visited: A, B, C, D, E, I, L, M, N



BFS Traversal

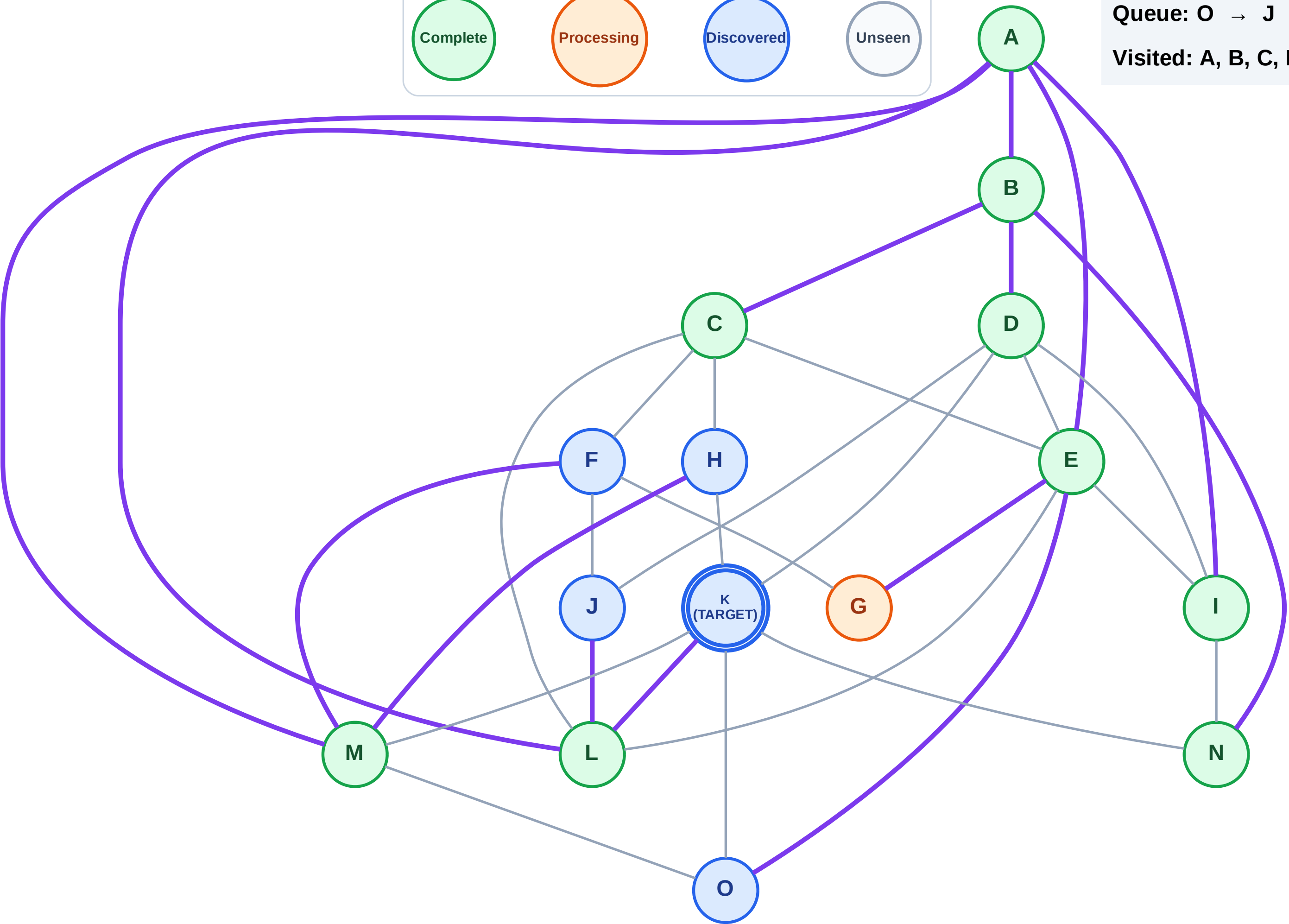
Step 20: Process node G

Legend



Queue: O → J → K → F → H

Visited: A, B, C, D, E, I, L, M, N



BFS Traversal

Step 21: No new neighbors from G

Legend

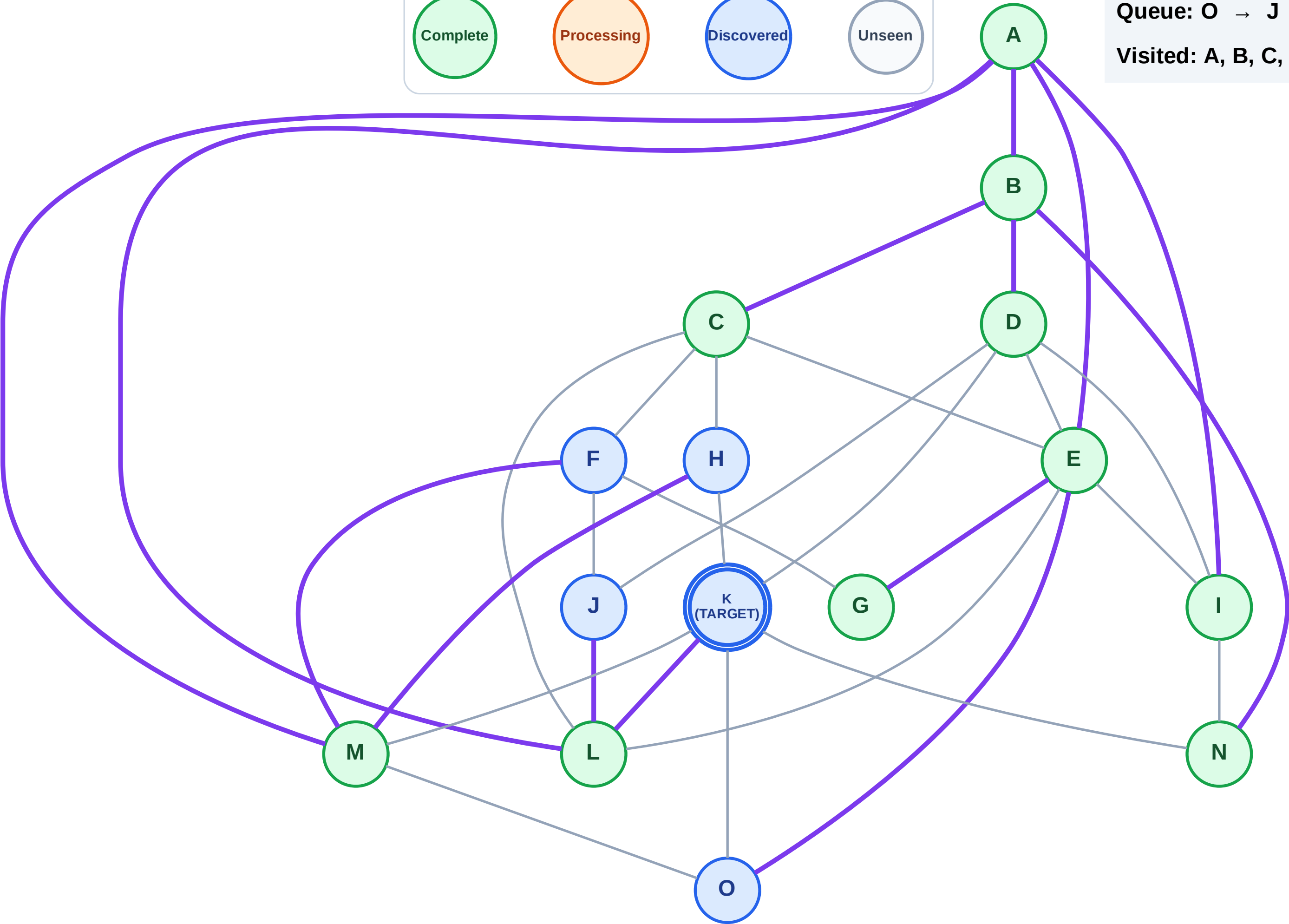
Complete

Processing

Discovered

Unseen

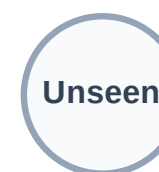
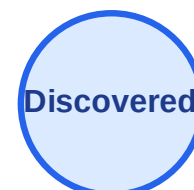
Queue: O → J → K → F → H
Visited: A, B, C, D, E, G, I, L, M, N



BFS Traversal

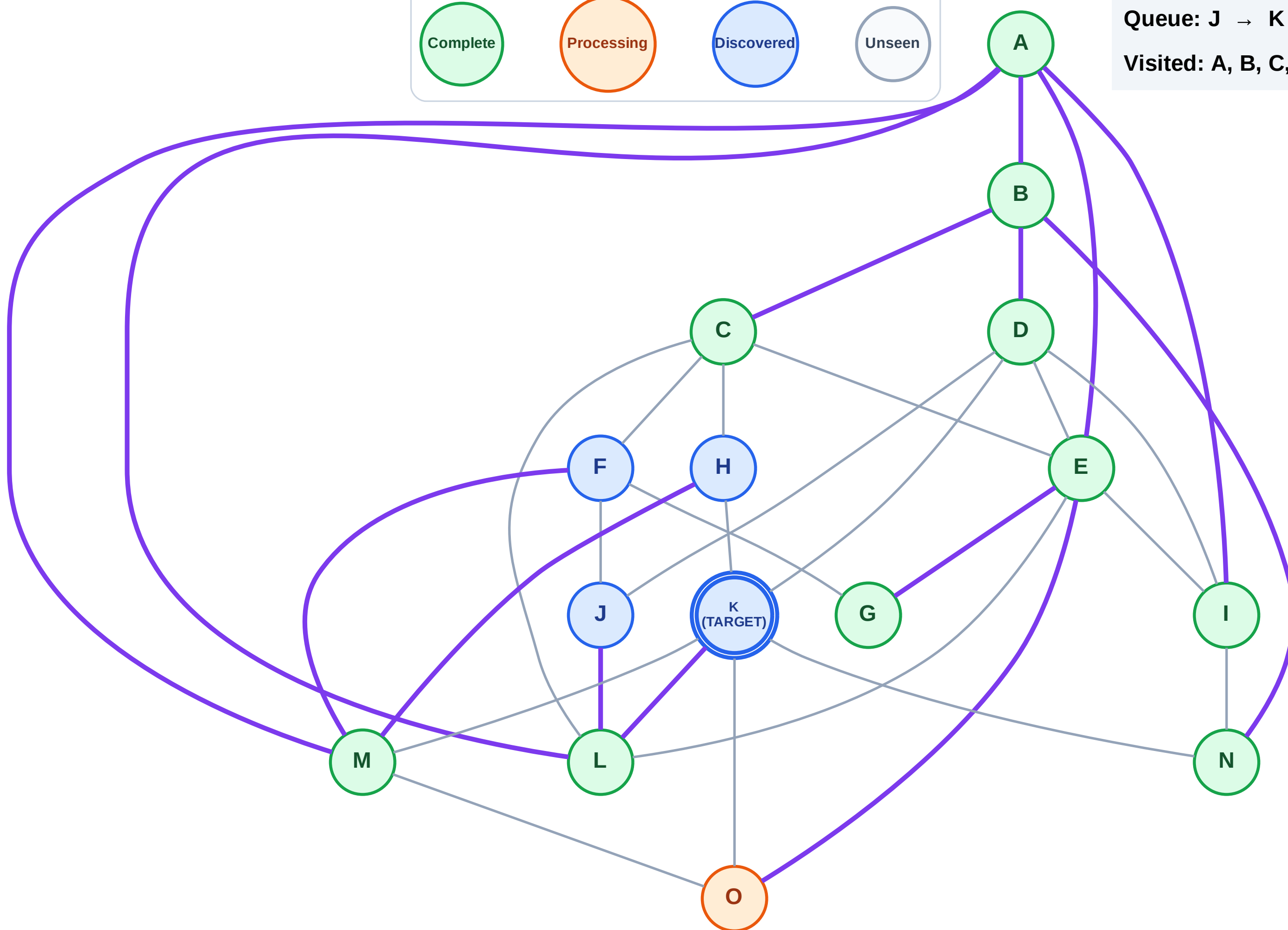
Step 22: Process node O

Legend



Queue: J → K → F → H

Visited: A, B, C, D, E, G, I, L, M, N



BFS Traversal

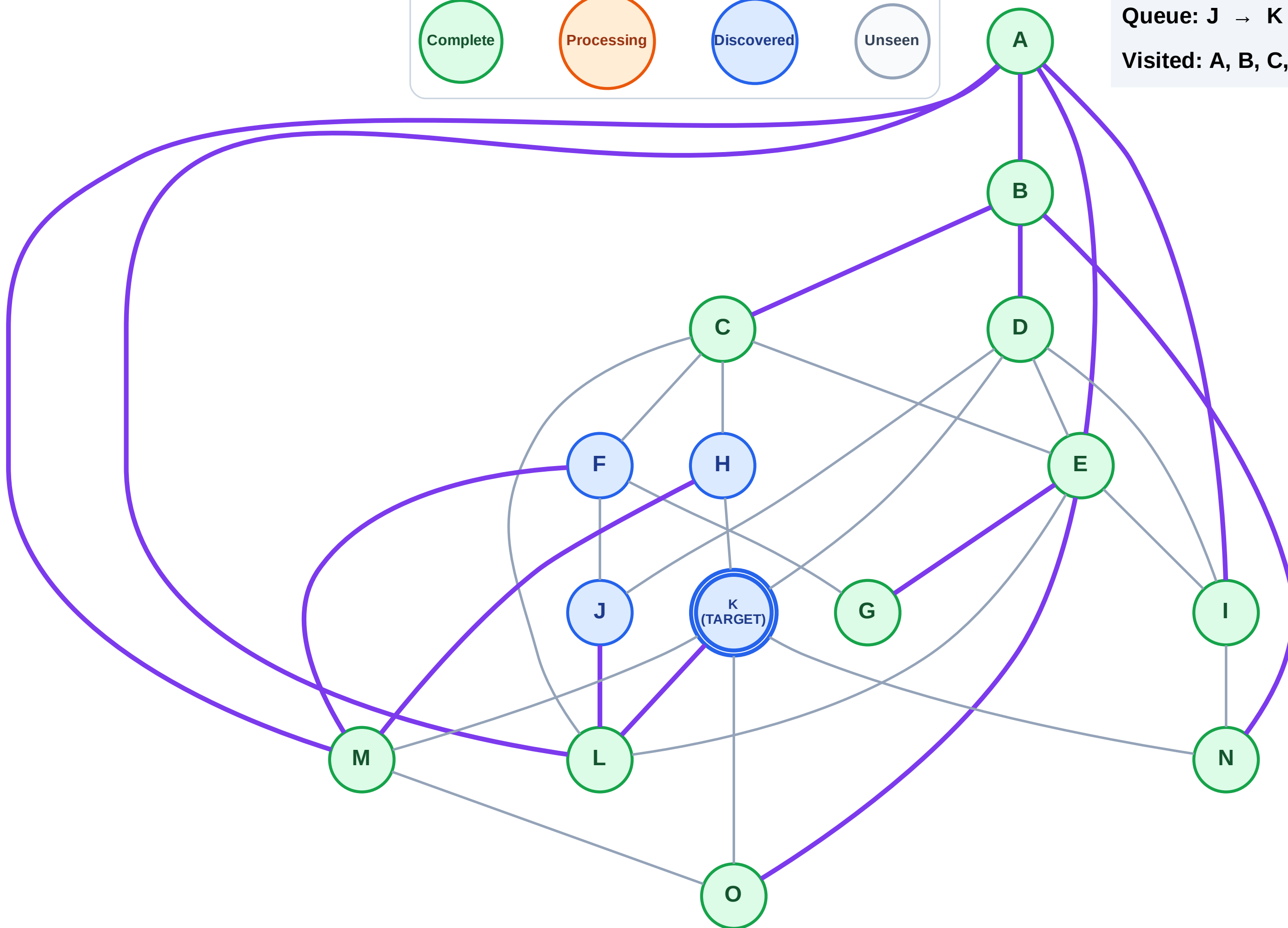
Step 23: No new neighbors from O

Legend



Queue: J → K → F → H

Visited: A, B, C, D, E, G, I, L, M, N, O



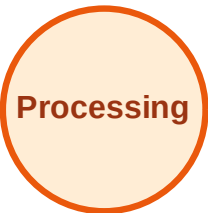
BFS Traversal

Step 24: Process node J

Legend



Complete



Processing



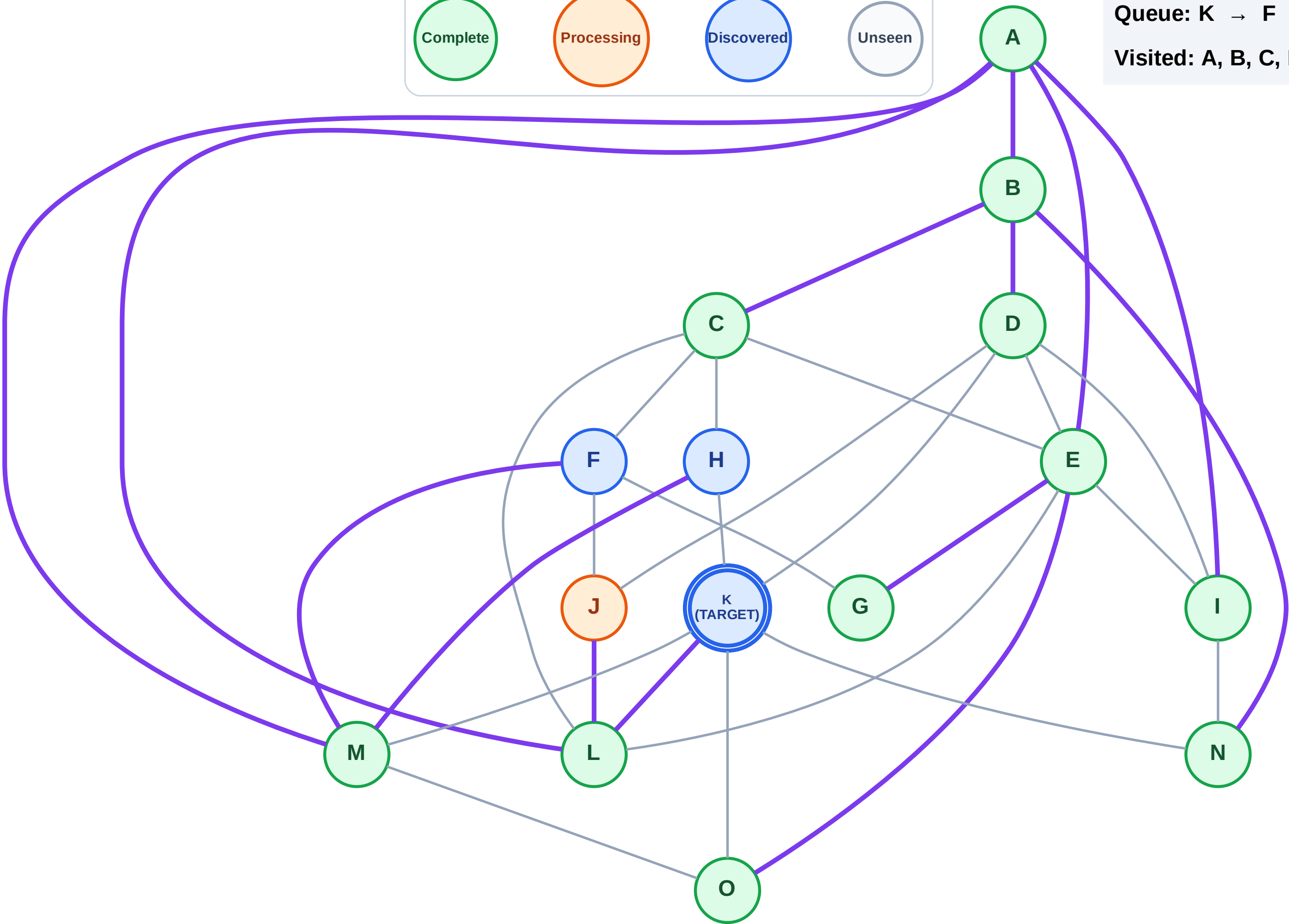
Discovered



Unseen

Queue: K → F → H

Visited: A, B, C, D, E, G, I, L, M, N, O



BFS Traversal

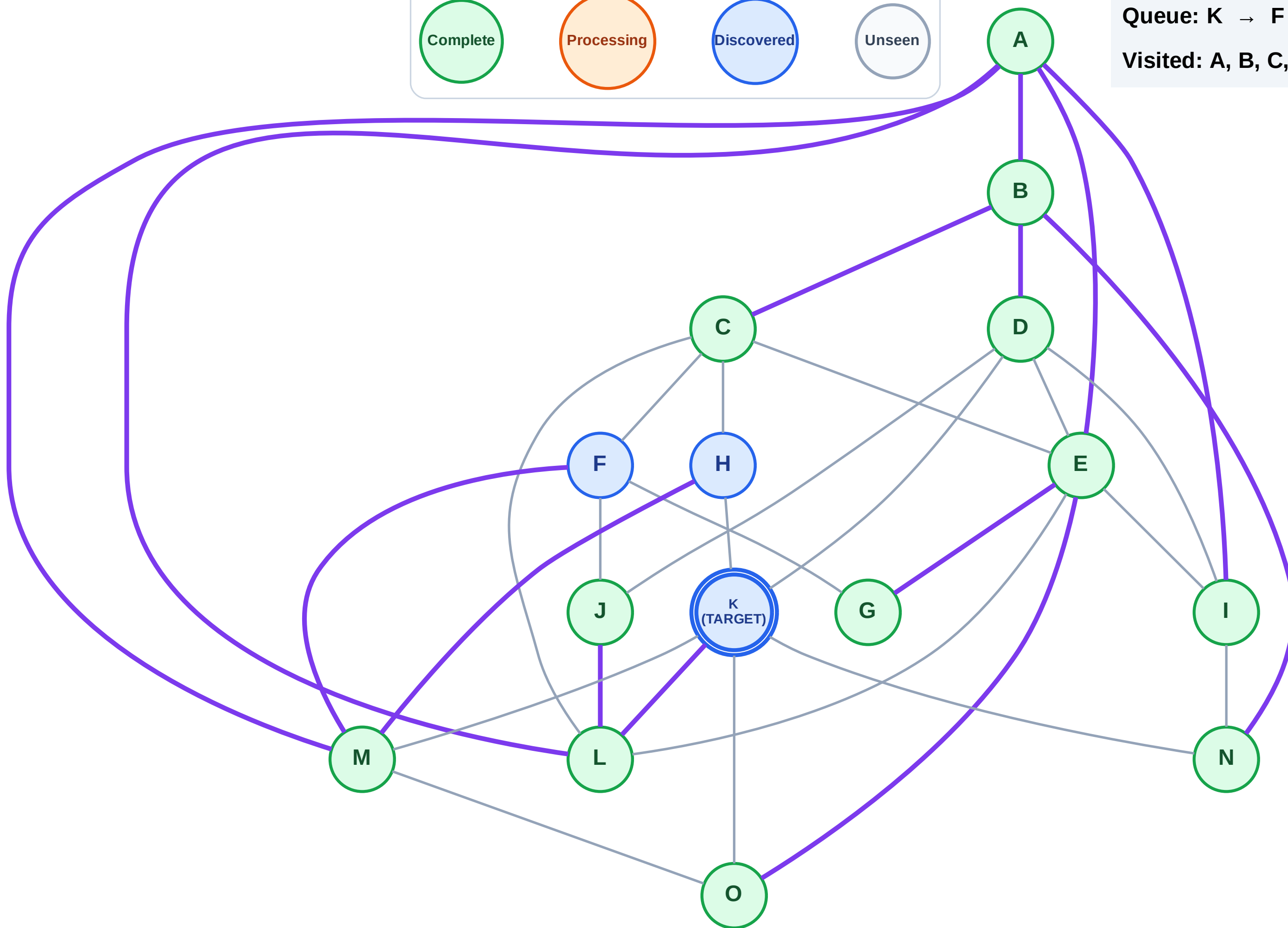
Step 25: No new neighbors from J

Legend



Queue: K → F → H

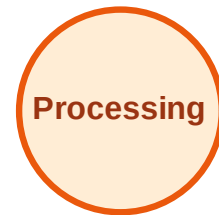
Visited: A, B, C, D, E, G, I, J, L, M, N, O



BFS Traversal

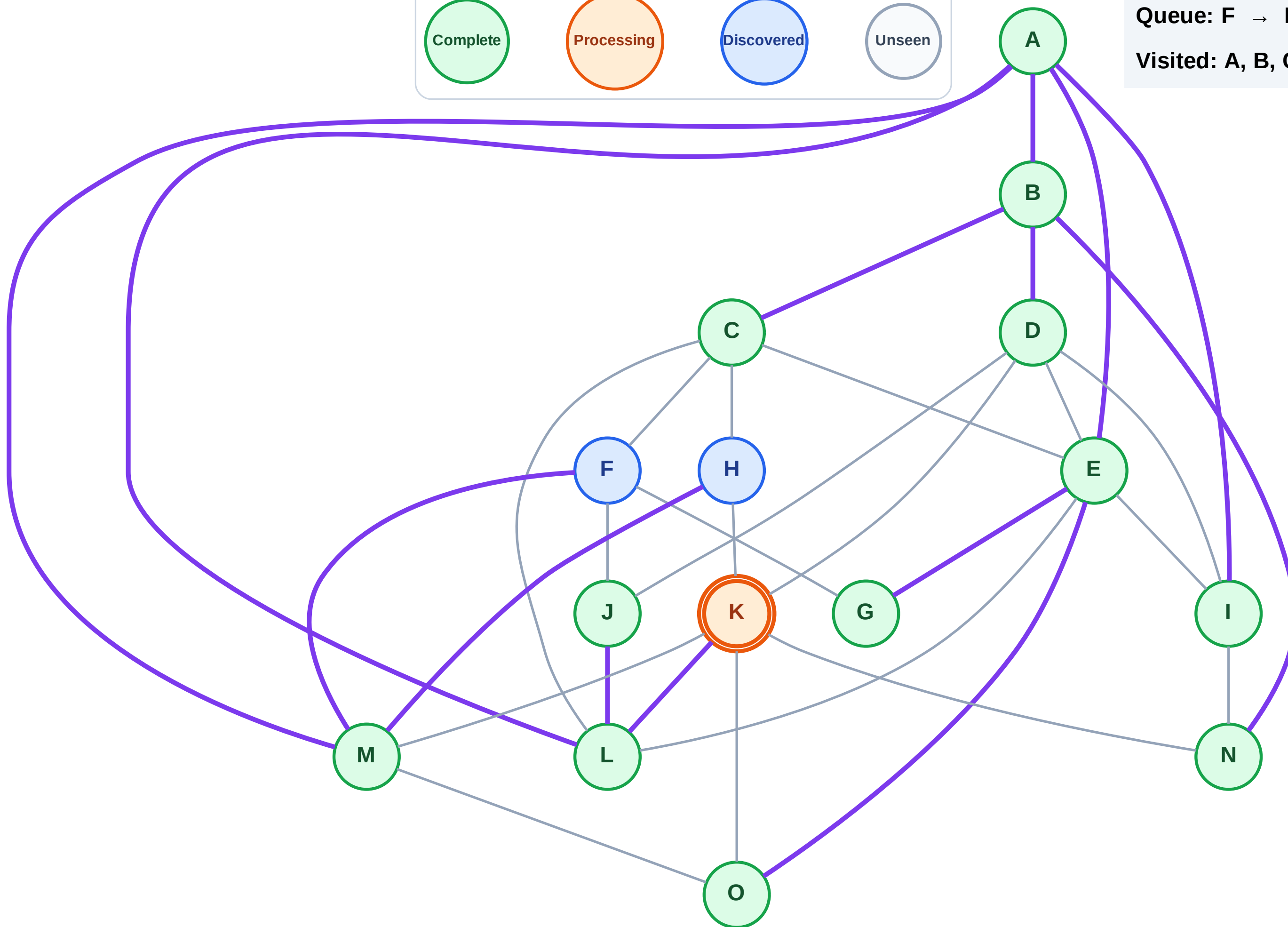
Step 26: Process node K

Legend



Queue: F → H

Visited: A, B, C, D, E, G, I, J, L, M, N, O



BFS Traversal

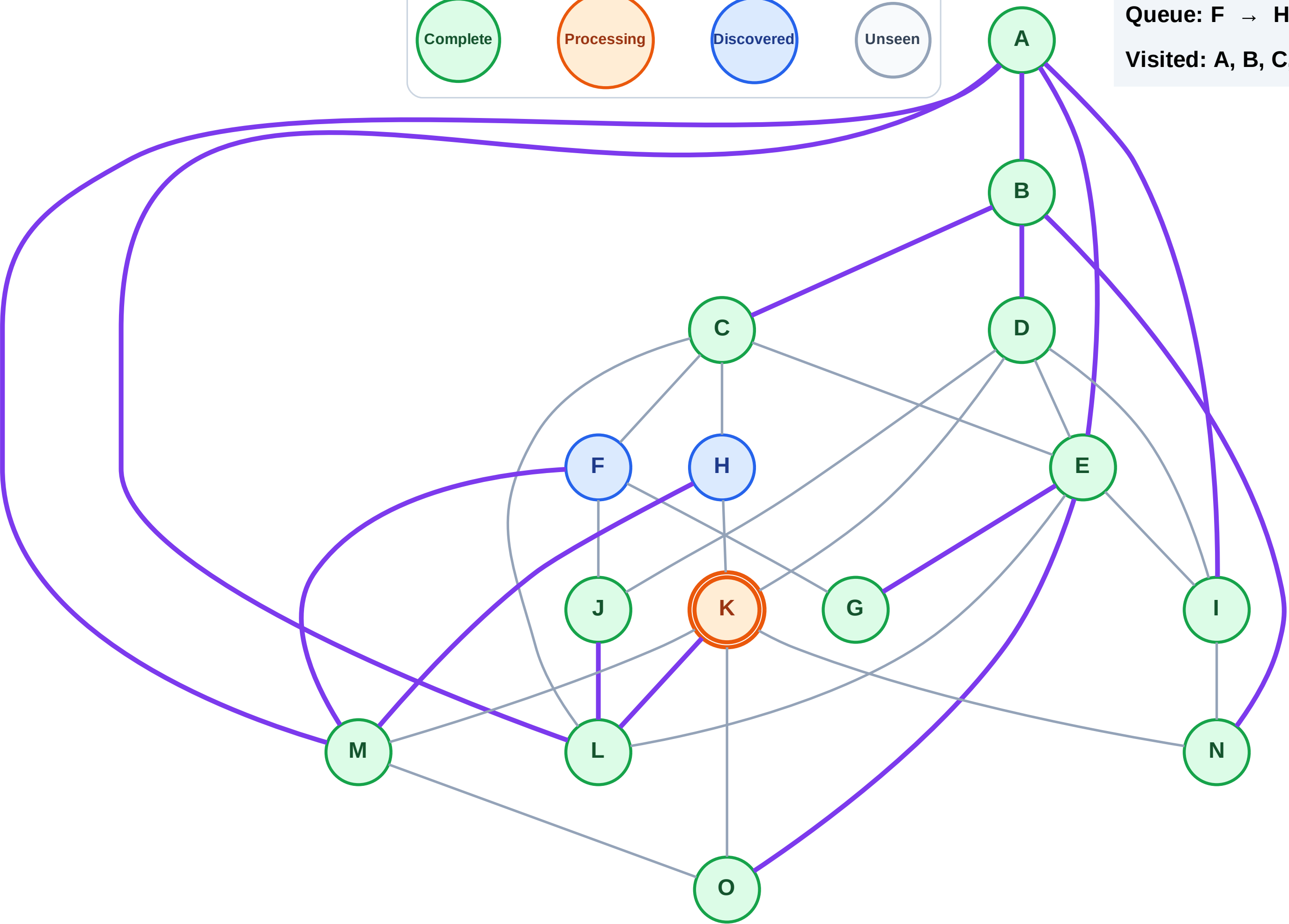
Step 27: TARGET NODE K REACHED! Path (2 hops): A → L → K

Legend



Queue: F → H

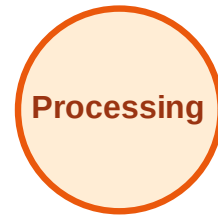
Visited: A, B, C, D, E, G, I, J, K, L, M, N, O



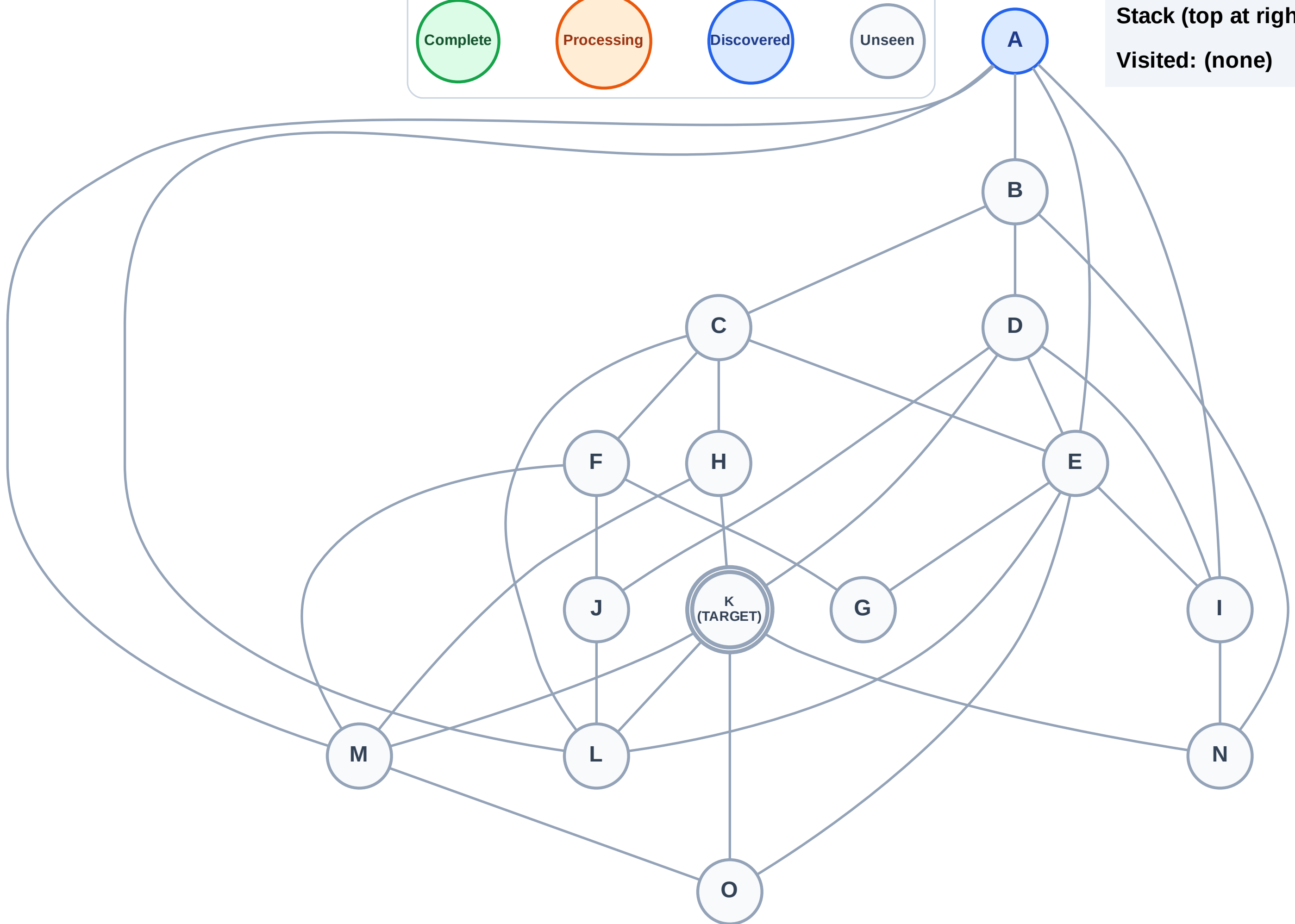
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



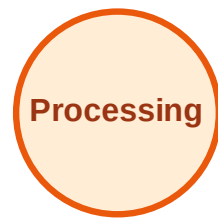
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



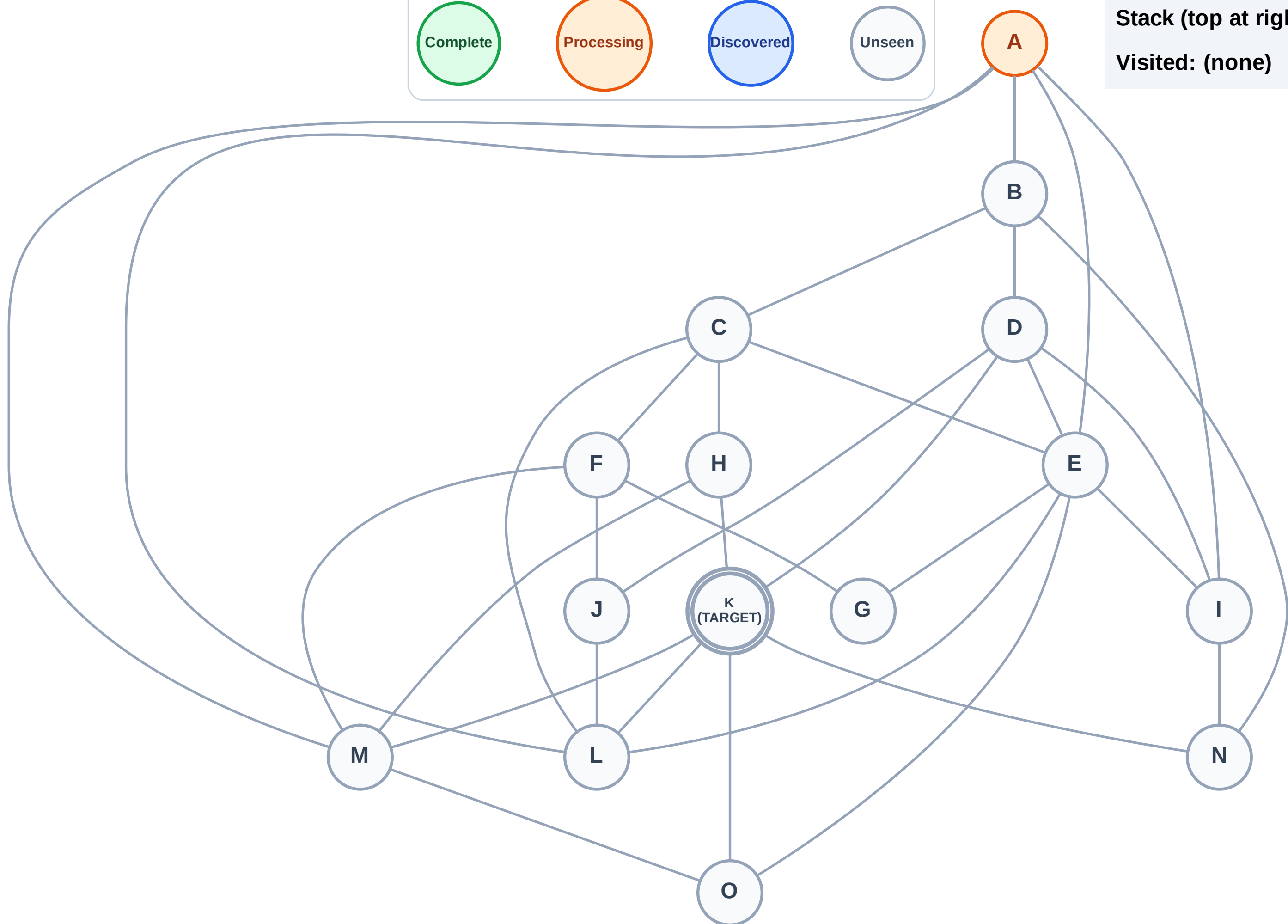
Discovered



Unseen

Stack (top at right): (empty)

Visited: (none)



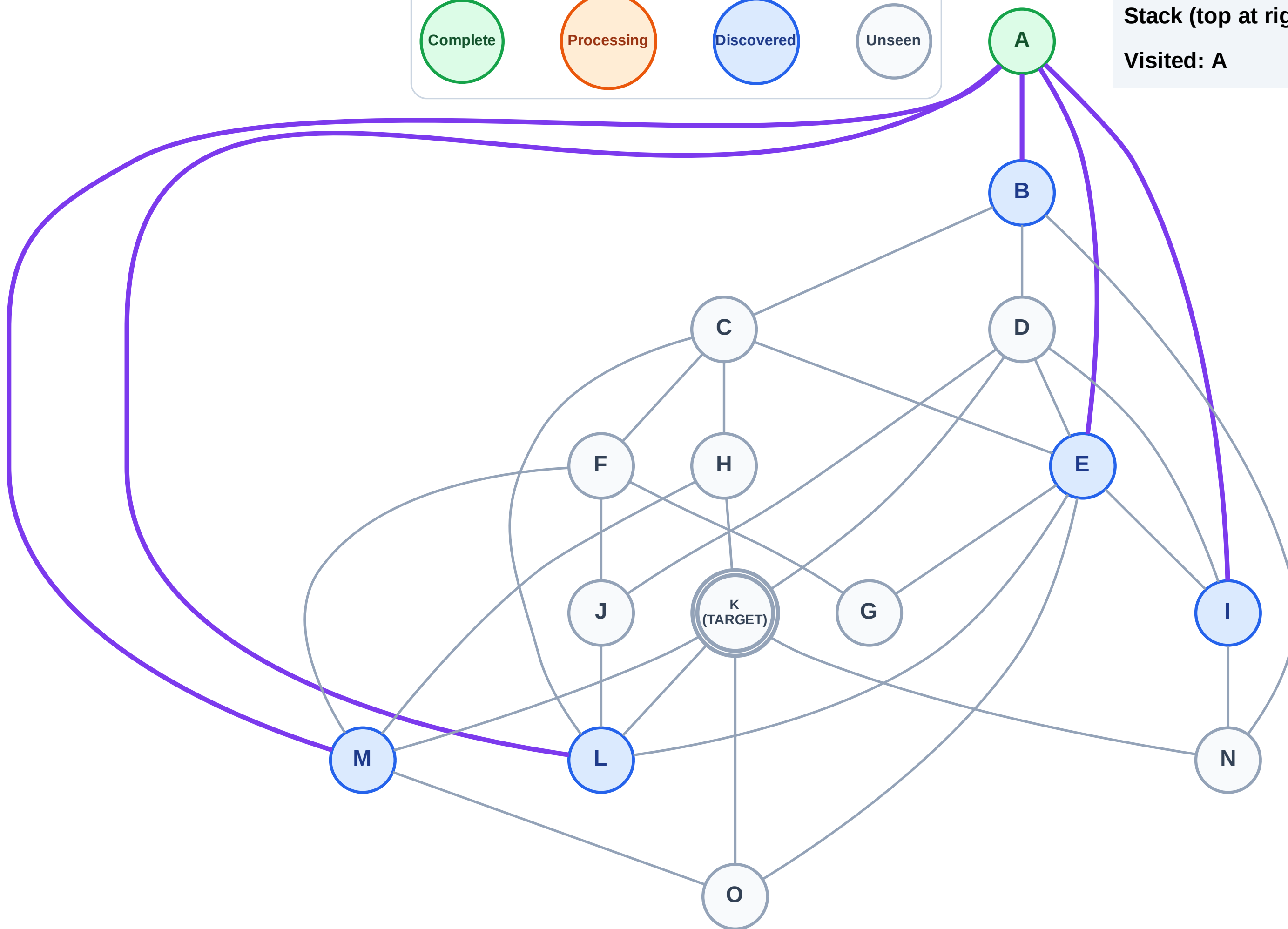
DFS Traversal

Step 3: Discover M, L, I, E, B from A

Legend



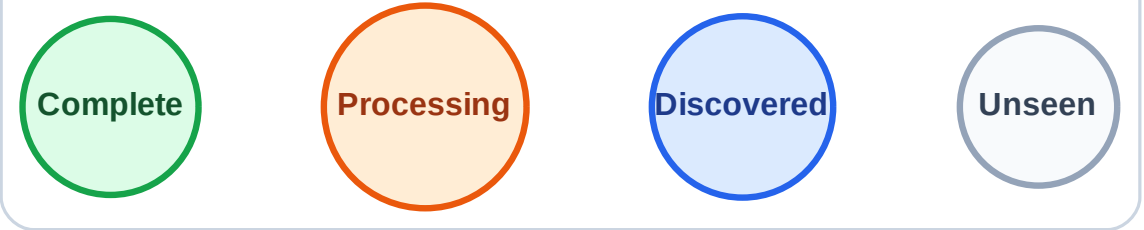
Stack (top at right): M | L | I | E | B
Visited: A



DFS Traversal

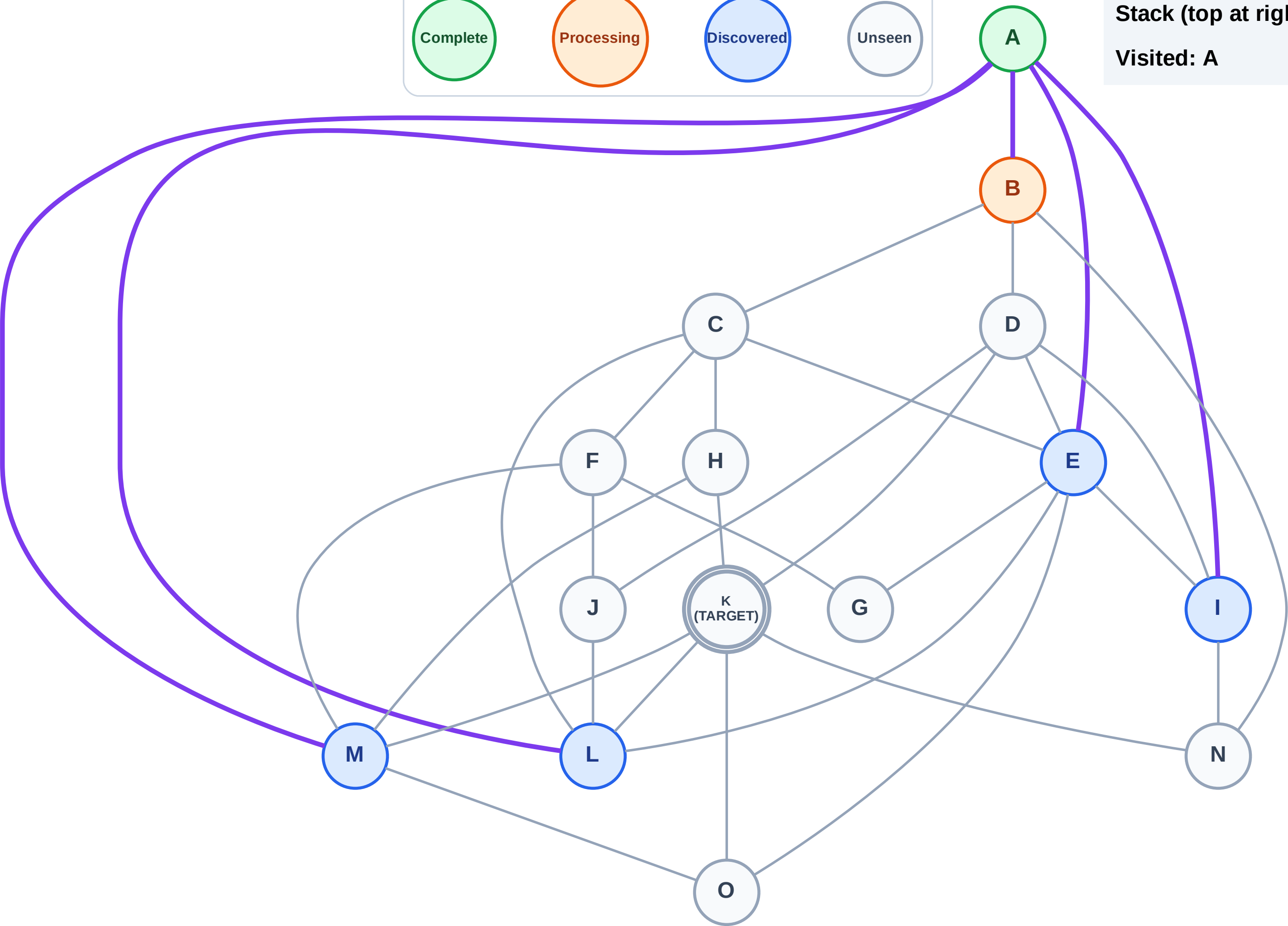
Step 4: Process node B

Legend



Stack (top at right): M | L | I | E

Visited: A



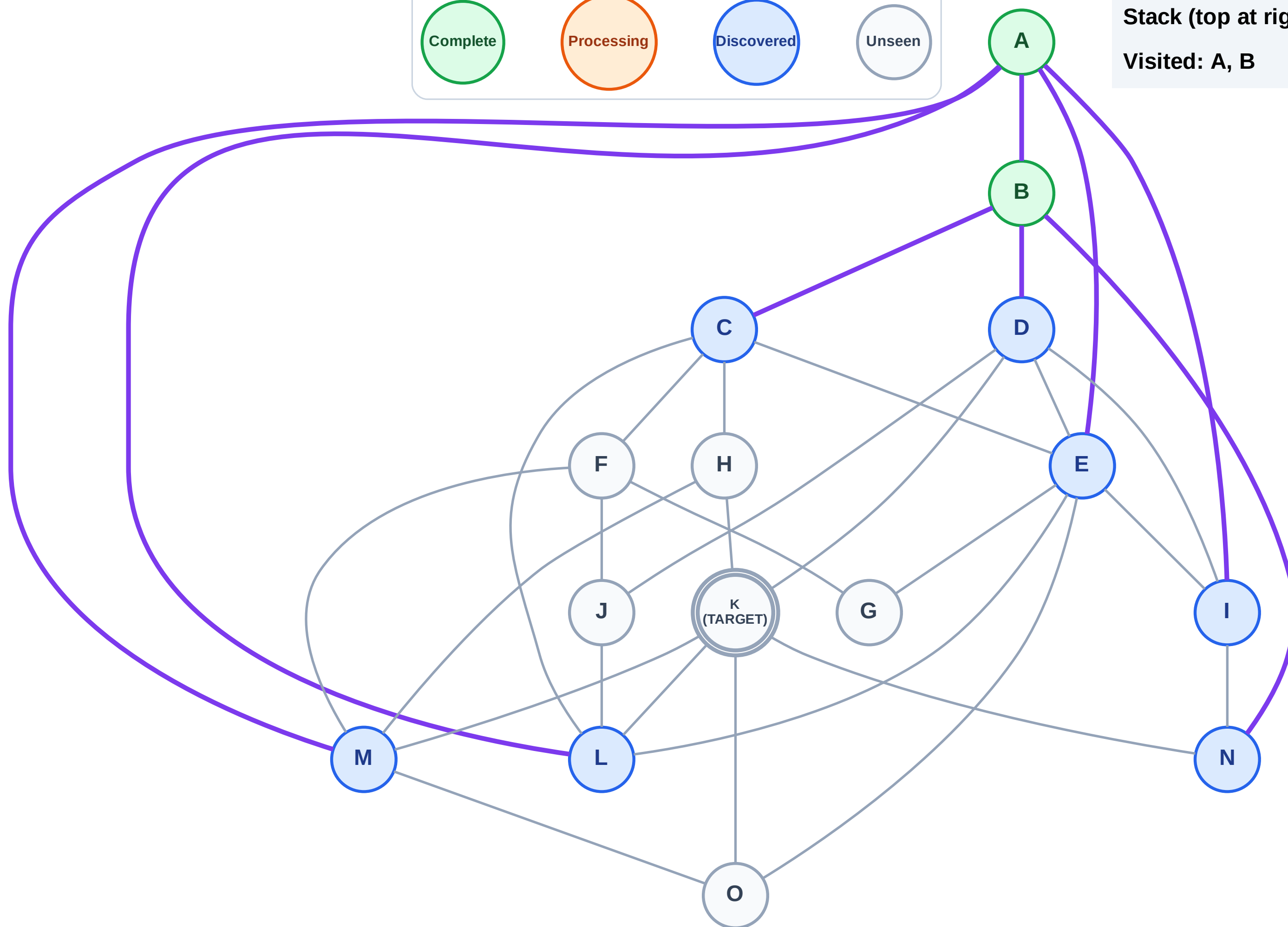
Step 5: Discover N, D, C from B

Legend



Complete Processing Discovered Unseen

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

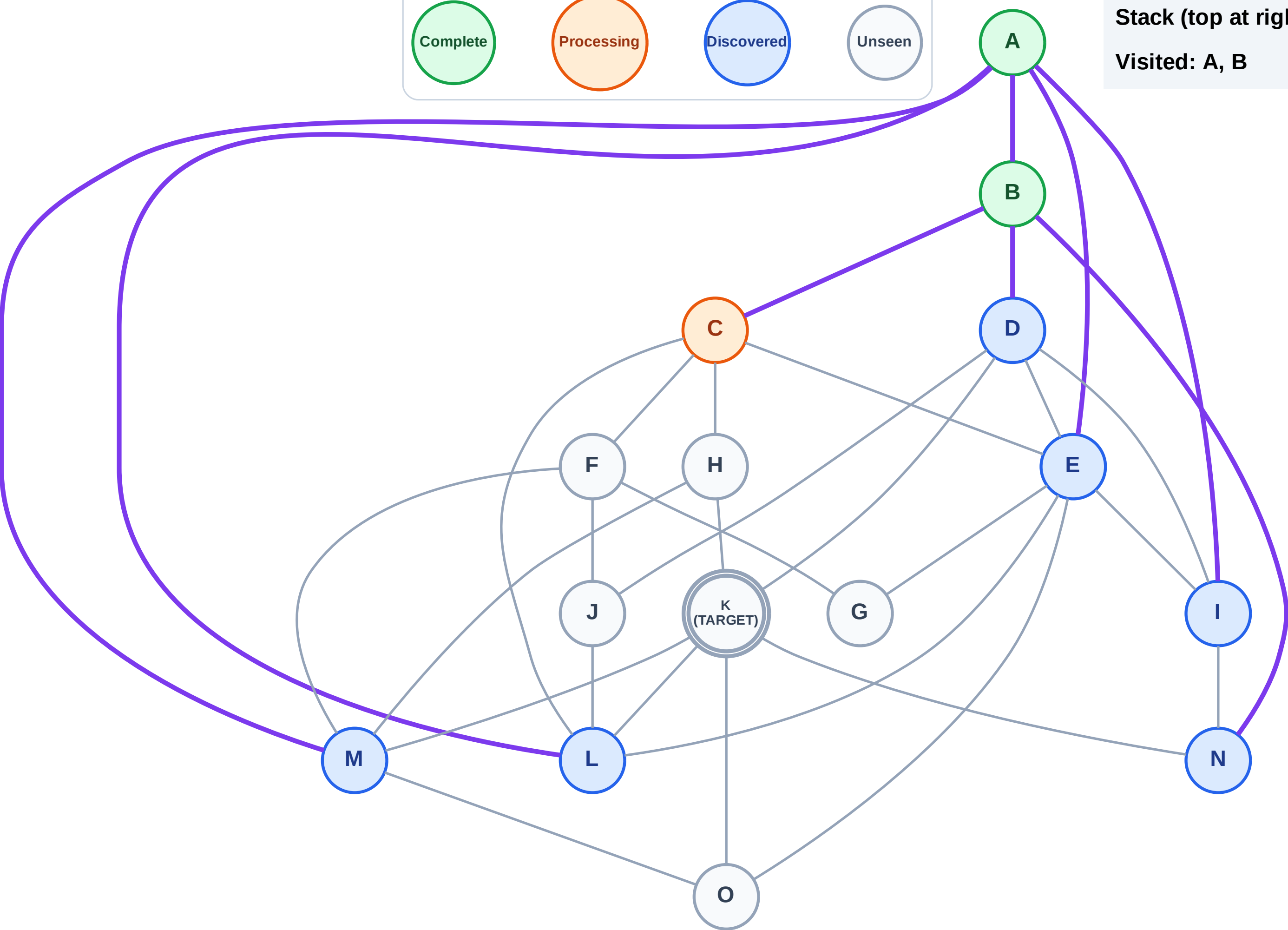
Processing

Discovered

Unseen

Stack (top at right): M | L | I | E | N | D

Visited: A, B



DFS Traversal

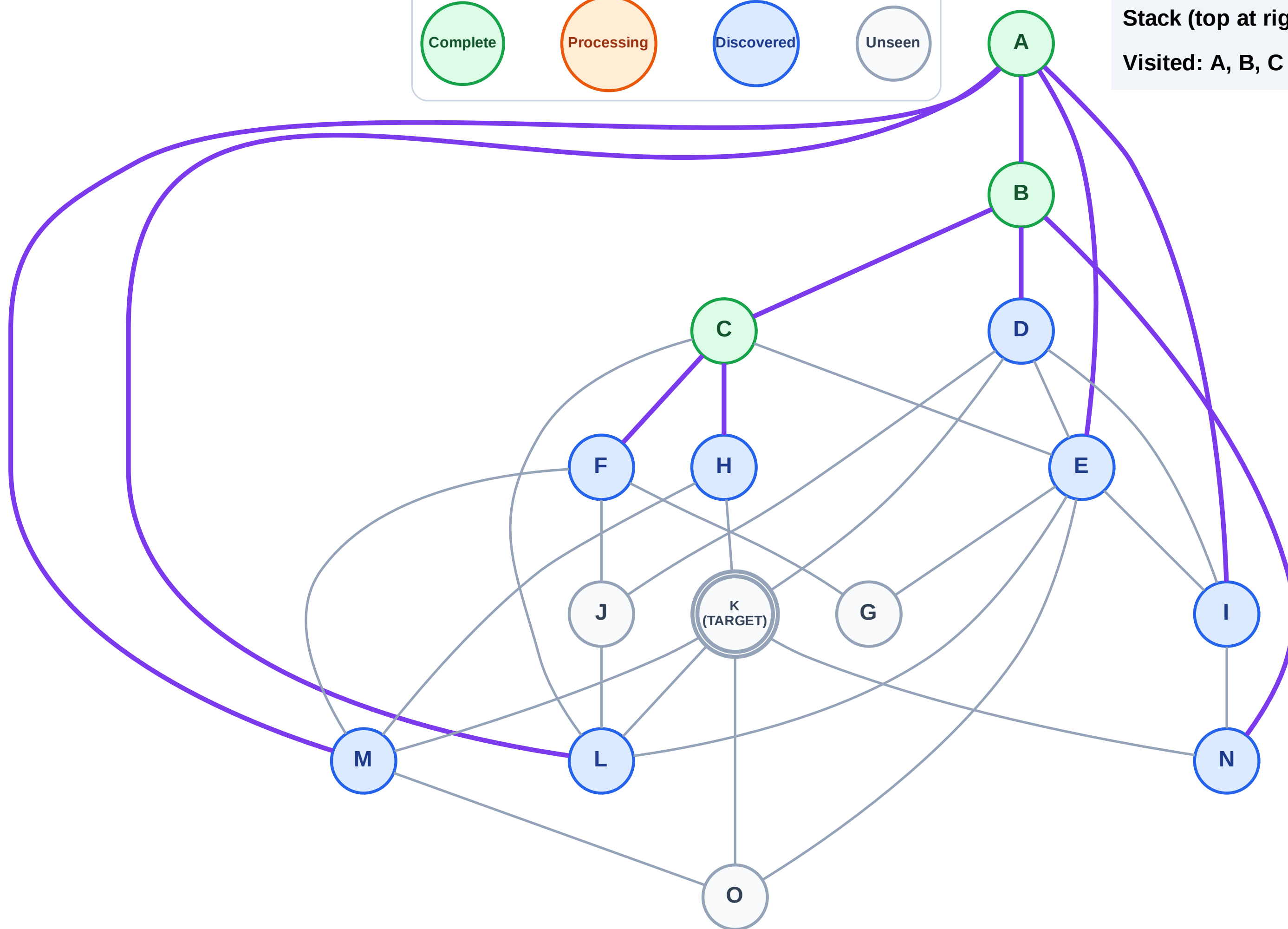
Step 7: Discover H, F from C

Legend



Stack (top at right): M | L | I | E | N | D | H | F

Visited: A, B, C



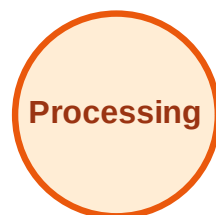
DFS Traversal

Step 8: Process node F

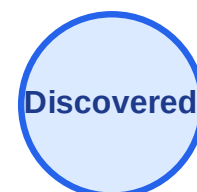
Legend



Complete



Processing



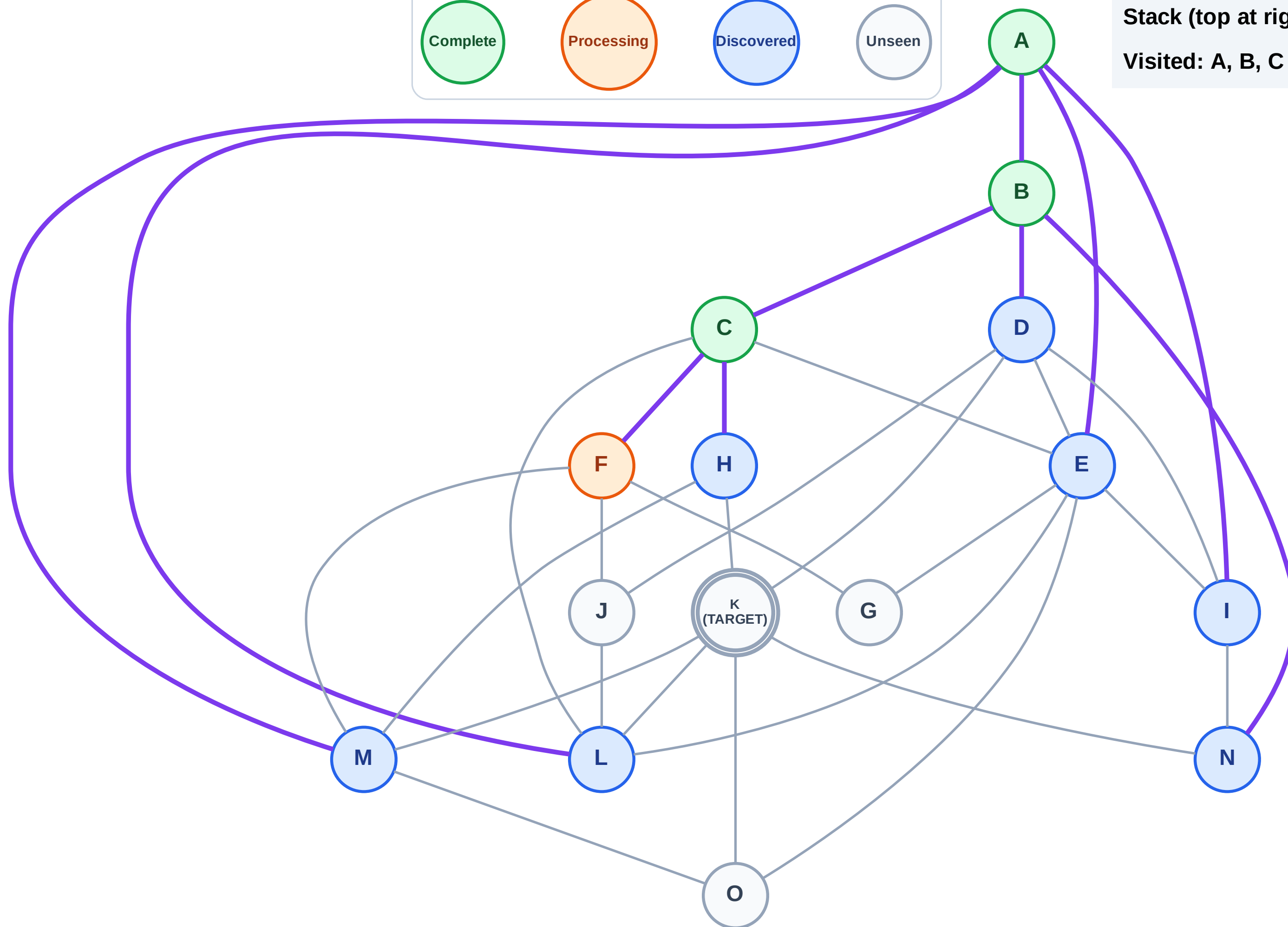
Discovered



Unseen

Stack (top at right): M | L | I | E | N | D | H

Visited: A, B, C



DFS Traversal

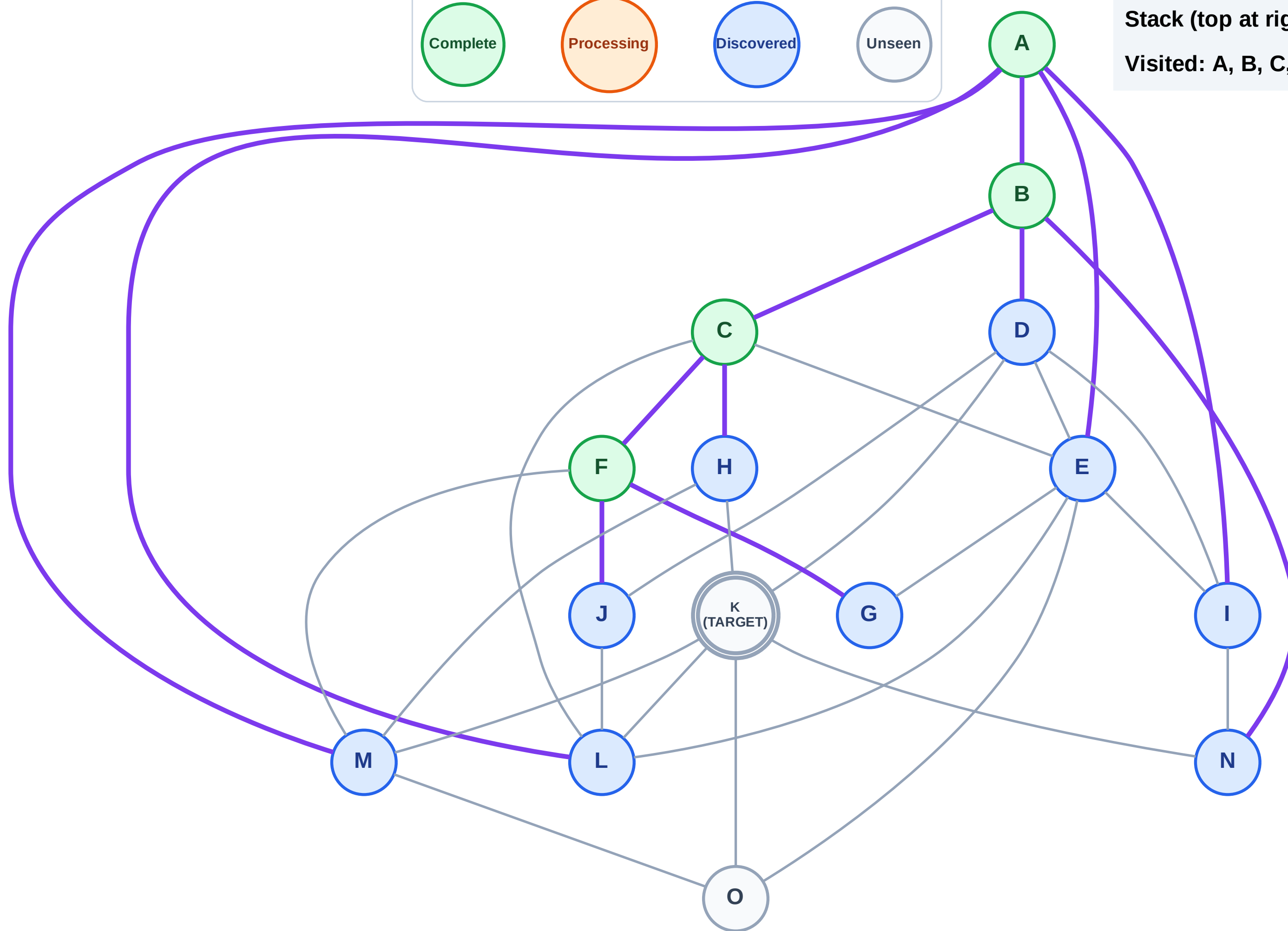
Step 9: Discover J, G from F

Legend



Stack (top at right): M | L | I | E | N | D | H | J | G

Visited: A, B, C, F



DFS Traversal

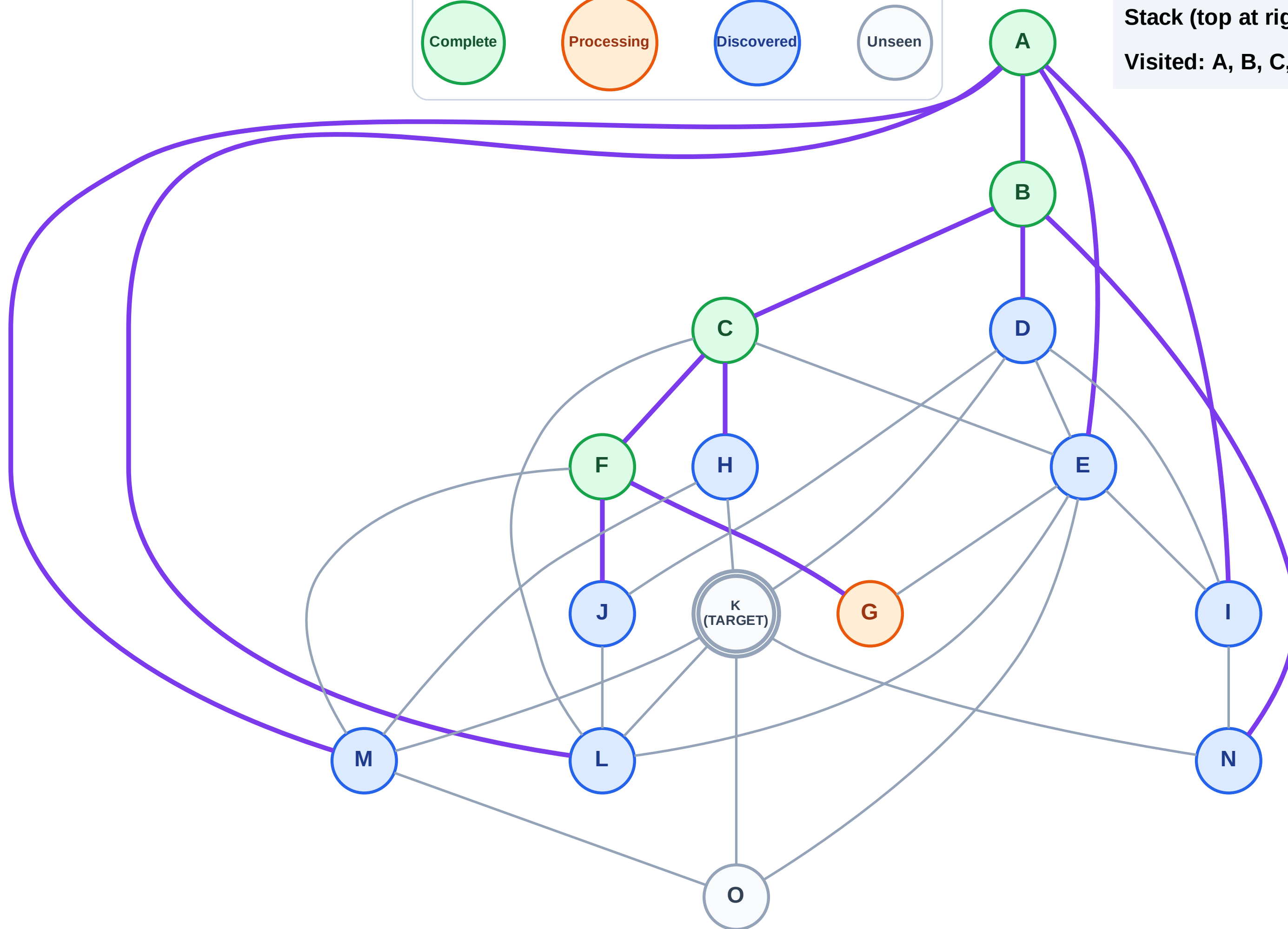
Step 10: Process node G

Legend



Stack (top at right): M | L | I | E | N | D | H | J

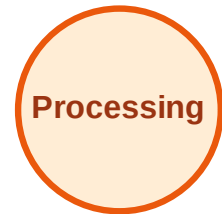
Visited: A, B, C, F



DFS Traversal

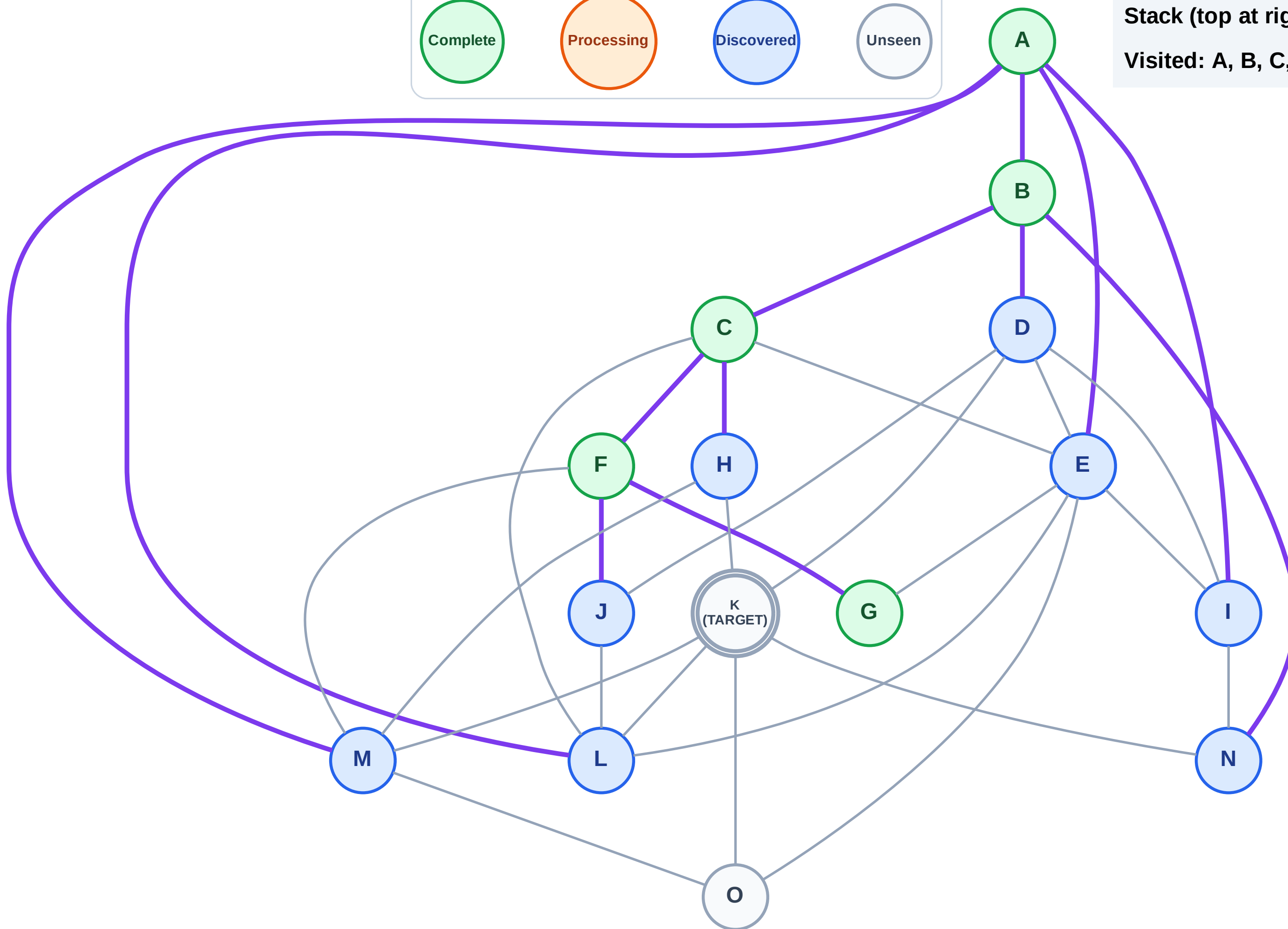
Step 11: No new neighbors from G

Legend



Stack (top at right): M | L | I | E | N | D | H | J

Visited: A, B, C, F, G



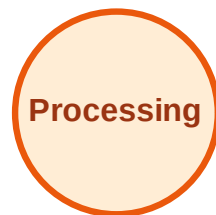
DFS Traversal

Step 12: Process node J

Legend



Complete



Processing



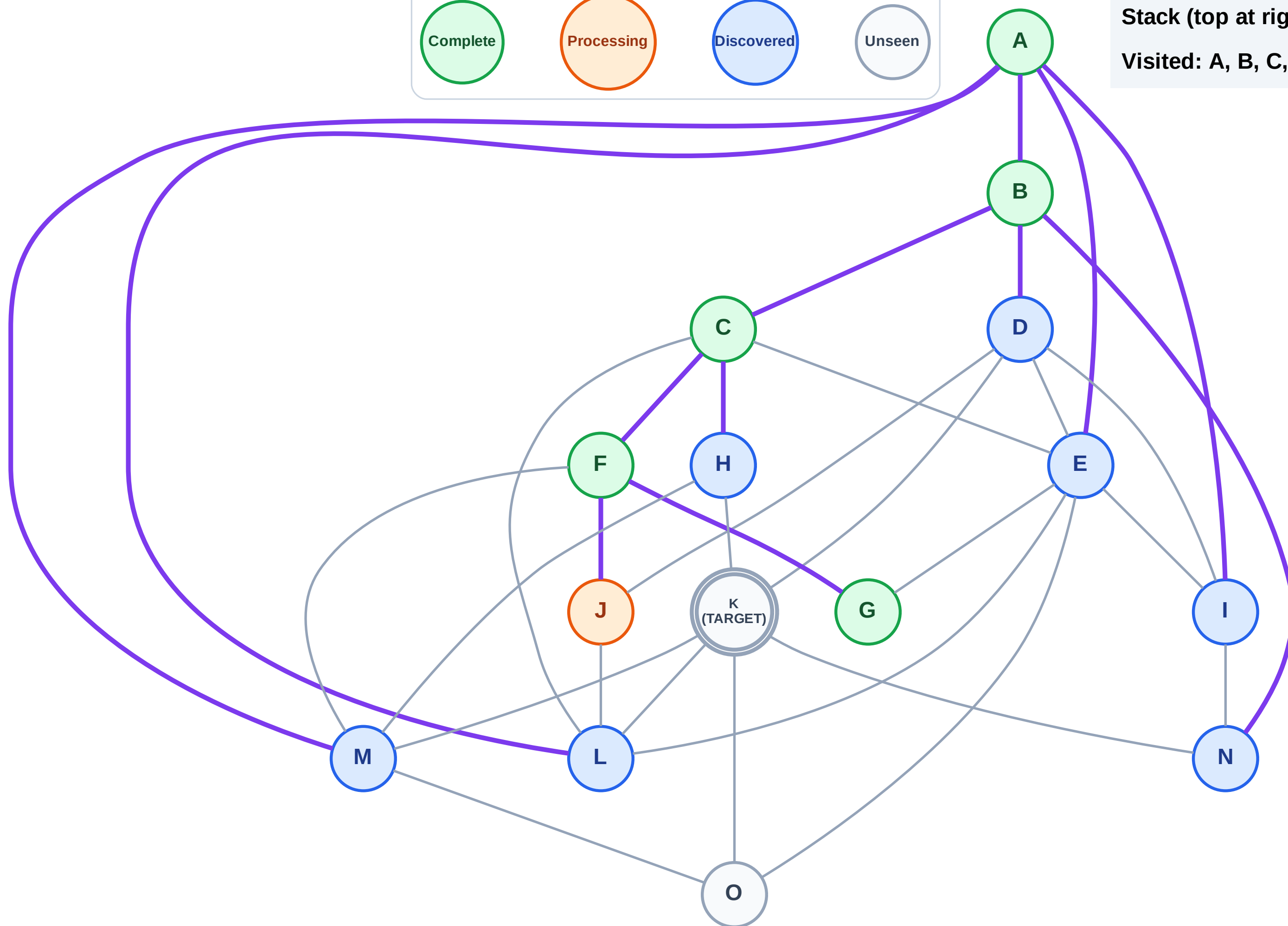
Discovered



Unseen

Stack (top at right): M | L | I | E | N | D | H

Visited: A, B, C, F, G



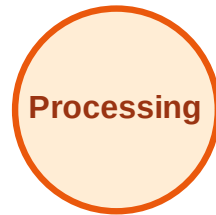
DFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



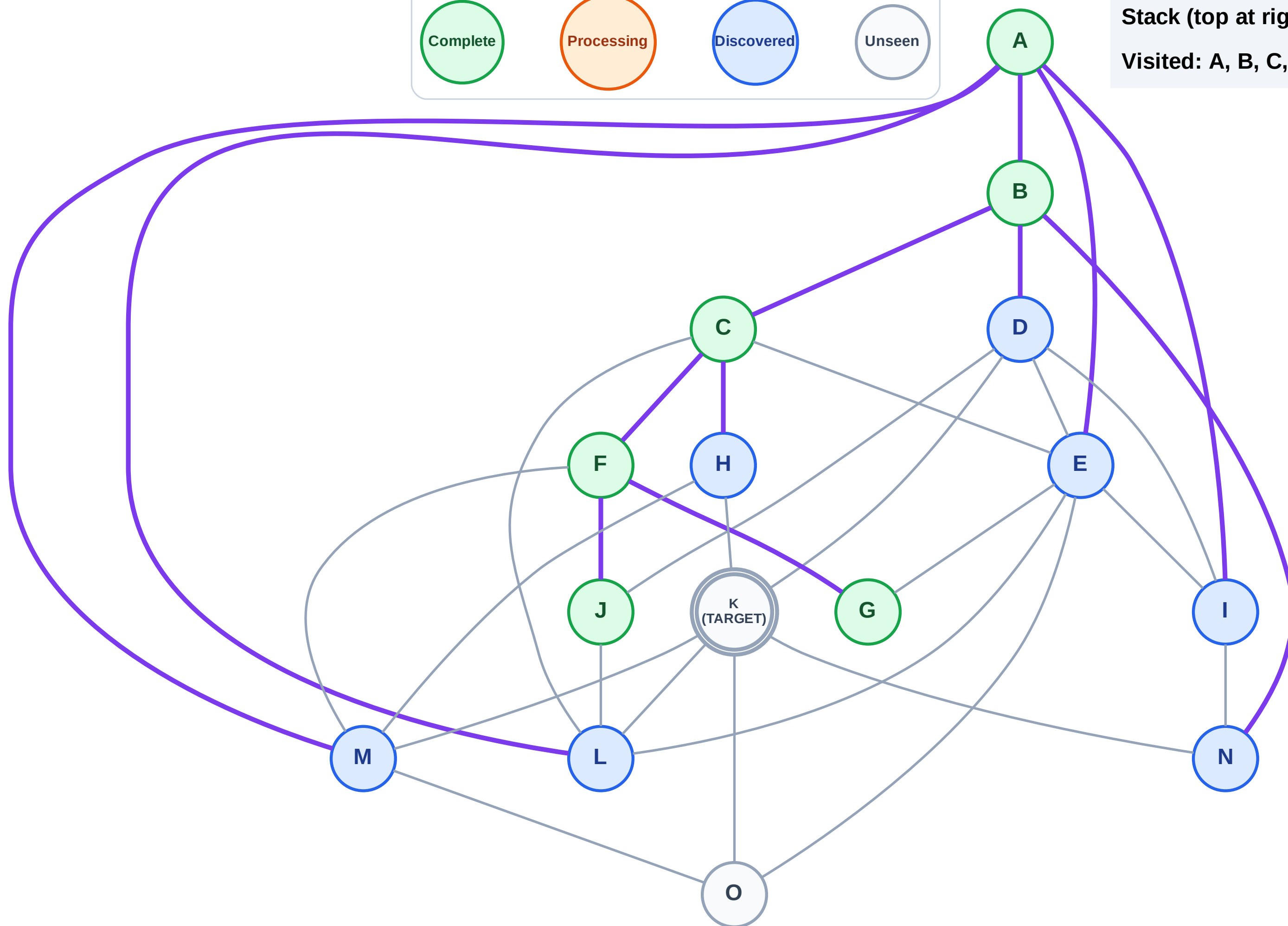
Discovered



Unseen

Stack (top at right): M | L | I | E | N | D | H

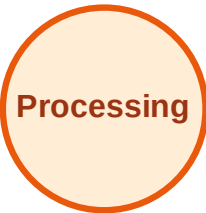
Visited: A, B, C, F, G, J



DFS Traversal

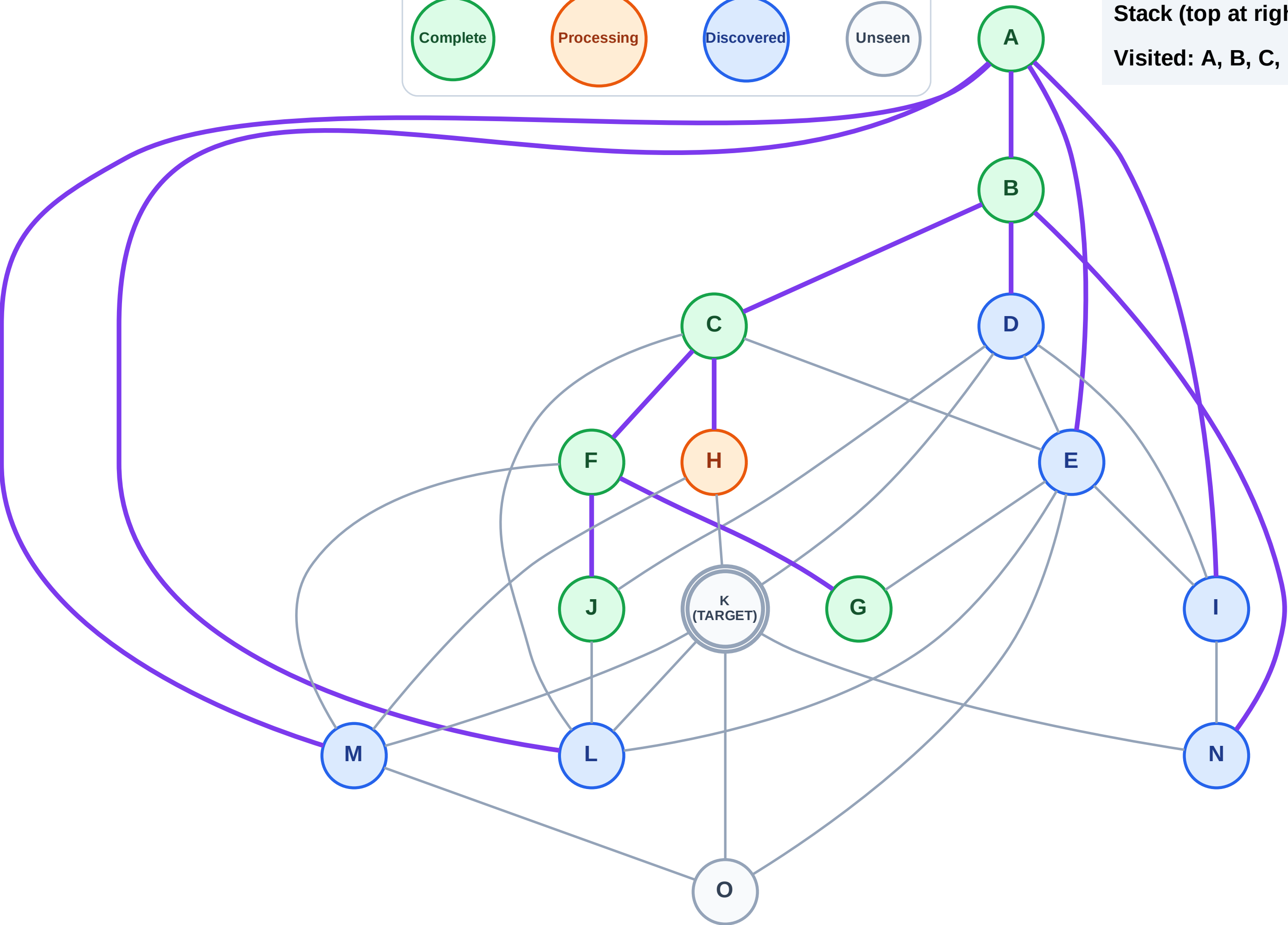
Step 14: Process node H

Legend



Stack (top at right): M | L | I | E | N | D

Visited: A, B, C, F, G, J



DFS Traversal

Step 15: Discover K from H

Legend

Complete

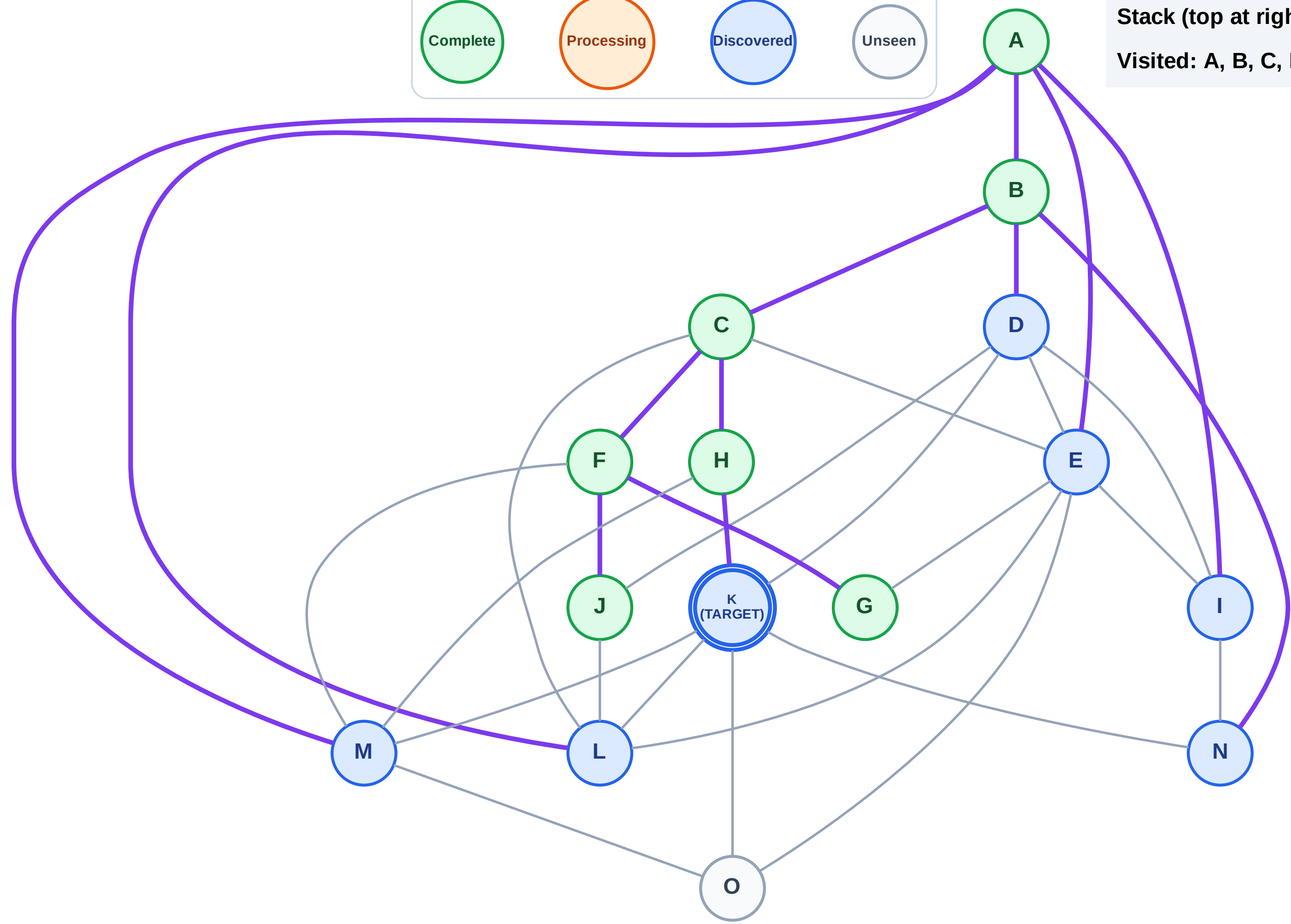
Processing

Discovered

Unseen

Stack (top at right): M | L | I | E | N | D | K

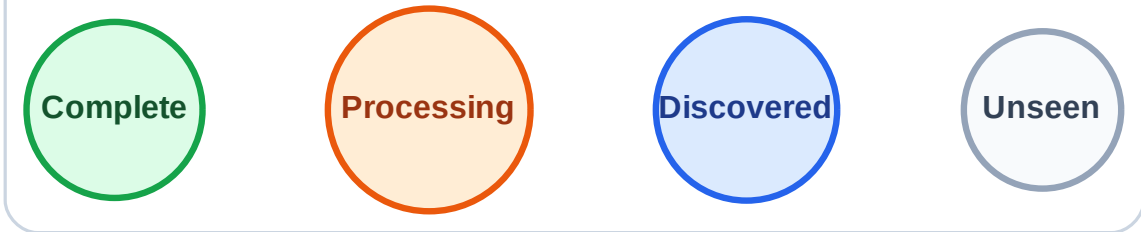
Visited: A, B, C, F, G, H, J



DFS Traversal

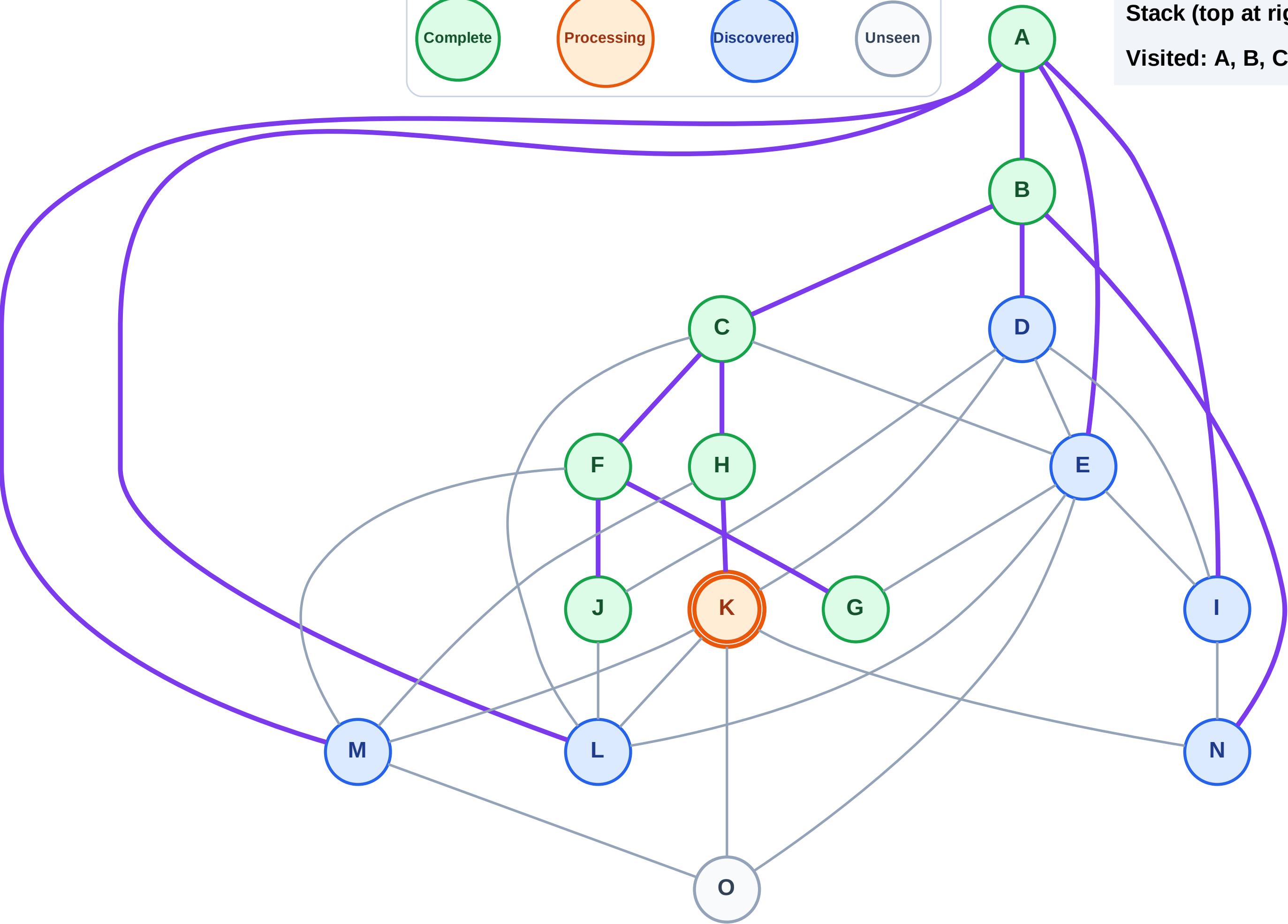
Step 16: Process node K

Legend



Stack (top at right): M | L | I | E | N | D

Visited: A, B, C, F, G, H, J



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (4 hops): A → B → C → H → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | I | E | N | D

Visited: A, B, C, F, G, H, J, K

